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The Emotion Engine

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Abstract

Language inevitably compresses and simplifies human emotion, causing subtle affective nuances to be lost in verbal expression. Although large language models (LLMs) can recognise and generate emotional language, most existing methods rely on discrete emotion labels or token-level control, which limits fine-grained and intuitive affective manipulation.

This project proposes an Emotion Engine that represents emotion as a continuous latent vector, rather than explicit linguistic categories. Given an input text, the system extracts its underlying emotional state as a low-dimensional emotion vector, which composes multiple overlapping affective directions. Users can directly adjust intensities of those vectors before regenerating text, that reflects the edited emotional state. Importantly, this operation is performed prior to language generation, bypassing the lossy compression inherent in verbalisation.

By leveraging recent advances in latent-space reasoning and internal activation steering in LLMs, the Emotion Engine enables controllable, pre-verbal emotion editing within a text-to-text pipeline. Human communication is always mediated by media. This project can affect all communication in the sense that it deals with feelings evoked by the media. Recent creative act with LLM prompt engineering may in fact be a role reversal, since creativity should serve before the verbalisation. By making it possible to edit the feelings that were really wanted to be conveyed, this project could bring about a more precise communication that society has never seen before.

Acknowledgments

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Contents

Chapter 1

Introduction

1.1 Words draw boundary lines across the ocean of "concepts"

When I was still in elementary school, my favourite biography in the library was Helen Keller's. Stricken by illness as a child, she lost both sight and hearing and bore the "triple hardship" of being blind, deaf, and mute, yet devoted herself to the advancement of education and welfare for the disadvantaged.

But Helen Keller's story is not merely a tale of "overcoming adversity." What is etched most vividly in my memory is the moment when she first understood the word "water." At first, Helen insisted that "cup" and "the water in it" were the same thing. However, the instant when Anne Sullivan signed "W-A-T-E-R" into her hand while letting the cold water from a pump run across Helen's palm, her world changed. When she could separate the object that flows from the container, the continuum of chaos was reshaped into an ordered world with partitions.

This scene exemplifies the essence of language. We do not grasp the external world directly as "things-in-themselves." The information flowing into our senses is, in truth, a continuum; language draws borders there and carves out units of meaning. Even the boundary between river and sea is not sharply inscribed by nature itself; it is defined for human convenience and by culture – "this far is river," "from here is sea." Words do not merely copy lines already inscribed upon the world; they are the very act of drawing those lines as needed. In other words, language is not just a label: it is the activity by which we impose boundaries within our cognition and shape the world we see. To learn words is not merely to expand vocabulary; it is to gain a way of partitioning the ocean of concepts and choosing how to look at the world. In that sense, we are, day by day, sketching new boundaries.

1.2 Verbalisation is a lossy compression

Verbalisation is an act of forcing the complex and continuous sensation into discretised vocabulary and grammar. When we perform **verbalisation**, there lurks a commonly overlooked risk; it is intrinsically **lossy**. The moment we verbalise, the fine nuances contained within raw feeling are abstracted into the fences of cognition that humanity has drawn at the level of words. Even if we choose the same term, the warmth that term carries depends on each person's embodied history, and it is eventually averaged out. The relief or pride we felt when we opened our old school yearbook for the first time in a decade, once put into text, may be flattened into a bland "how nostalgic." Moreover, lexicons and meanings for emotions differ greatly across languages and cultures[1]; terms like "anger" or "joy" do not necessarily point to the same nuances around the world. And yet humanity cannot let go of language. Even if it is a lossy compression, language remains the only standardised interface that connects the human heart to society.

1.3 Objectives

The goal of this project is not to deny language but to design a parallel circuit that supplements the components lost in the compression of verbalisation, i.e. a circuit for restoring what words leave behind. What if we could scoop up the human central cognition directly, without passing through the rendering pipeline of language, and translate it into other forms of expression? Could we transfer the lingering sensation that arises when reading a letter from your significant other into poetry or essays? Or could we improve novel translations by capturing emotions lost through differences in lexical definitions between languages? Here lies an uncharted gap in our understanding of human cognition.

This project aims to emulate a kind of human synaesthesia. In other words, we aim to map emotions across modalities. We will develop the foundational technology to extract, manipulate, and re-synthesise a **pre-verbal, intermediate representation of emotion**, and explore possible ways to steer the model output using those representations of emotion.

The final product of this project is a computational system capable of extracting, manipulating, and presenting the emotions contained in input text without having to go through the compression of verbalisation. The system focuses not on what the text superficially tells, but on the emotional state that existed immediately before the text was produced, and treats it as a continuous vector representation.

When a user enters any text, the system analyses the emotional nuances contained in that text and visualises them as a low-dimensional continuous vector. These emotion vectors are not discrete labels such as "joy" and "sadness," but are expressed as multiple overlapping emotional directions, thus preserving subtle mixed emotions and fluctuations that cannot be captured by language.

Users can intuitively manipulate this emotion code space. For example, by strengthening only a specific emotional direction and weakening another direction for the simultaneous presence of `relief` and `loneliness` in a certain sentence, it is possible to sensitively edit the emotional state that one really wanted to convey. It is important to note that this operation is not performed at the linguistic level, such as vocabulary selection or stylistic adjustment, but rather on the emotional expression itself before the language is generated.

The edited emotion code is again transformed into a linguistic expression. The text generated here is not simply a paraphrased sentence, but is the result of a re-expression that retains or intentionally transforms the emotion contained in the original sentence.

In this process, the user does not need to put their emotions into explicit words. The verbal averaging and simplification that is usually considered inevitable between emotion recognition and generation is bypassed, and the raw emotion felt by the user can be transferred to another form of expression in a more controllable manner.

This end product is neither a system for classifying and diagnosing emotions nor a tool for automatically generating emotional sentences. Rather, it is an interface that temporarily holds the emotions that people had before they put them into words as manipulatable intermediate expressions, from which new linguistic expressions can be launched. The ultimate value of this project lies in the fact that it makes it possible to return to the boundaries drawn by language and choose words again.

Chapter 2

Background

2.1 Emotion Representation in LLMs

2.1.1 Discrete Emotion Labels and Their Limitations

Early work treated emotion as a small set of discrete categories. The most influential taxonomy is Ekman's seven basic emotions: anger, contempt, disgust, enjoyment, fear, sadness, and surprise [2].

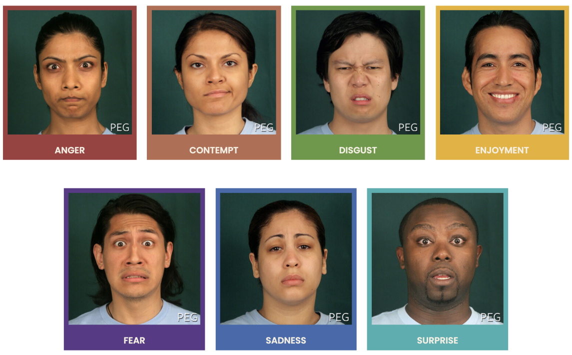

Figure 2.1: Paul Ekman's seven basic emotions [3]

This assumes that emotions are biologically universal and psychologically irreducible. This framework enabled early supervised emotion classification systems by providing a clear labeling scheme and facilitated the construction of benchmark datasets. However, the categorical assumption introduces strong inductive bias; emotions are treated as mutually exclusive and qualitatively distinct.

Subsequent datasets attempted to increase expressive granularity. GoEmotions [4], for example, expanded the label space to 27 emotion categories and allowed multi-label annotation.

Positive		Negative		Ambiguous
admiration 👏	joy 😃	anger 😡	grief 😢	confusion 😕
amusement 😂	love ❤️	annoyance 😒	nervousness 😬	curiosity 🤔
approval 👍	optimism 🤞	disappointment	remorse 😔	realization 💡
caring 🤗	pride 😌	disapproval 👎	sadness 😞	surprise 😲
desire 😍	relief 😅	disgust 🤮		
excitement 🤩		embarrassment 😳		
gratitude 🙏		fear 😨		

Figure 2.2: GoEmotions taxonomy: 28 emotion categories including "neutral". [5]

This enabled us to capture cases where multiple emotions occur in a single utterance. It is reported that emotions cluster by sentiment polarity and intensity, and that many categories exhibit high inter-label correlation. Nevertheless, despite its increased resolution, the dataset still relies on one-hot / sparse multi-hot encodings. Such encodings collapse emotional intensity and compositional structure into discrete indices, preventing the representation of graded transitions. For example, the difference between mild and intense anger cannot be represented, and the blend of emotions, such as bittersweet emotion, is also difficult.

Essentially, discrete labels force the learning problem into a classification regime that cannot capture continuity, geometry, or arithmetic structure. For example, two samples labeled `joy` may be emotionally distant, while `joy` and `contentment` may be emotionally close; yet this structure is invisible to the model itself. These limitations motivate a shift away from purely categorical representations toward continuous emotion spaces.

This problem of the discretised emotion labels is also raised by Wang and Zong (2021) [6]. They explicitly formalise this by arguing that one-hot emotion labels ignore latent relationships between emotion categories. They observe that emotion calsses are not orthogonal in human perception, but rather exhibit similarity in a gradation manner. More broadly, discrete labeling schemes prioritises the annotation convenience over the representational adequacy. While it simplify supervision and evaluation, they impose artificial bounadies that obscure the underlying structure of emotions. This mismatch becomes especially problematic when attempting cross-dataset transfer or cross-cultural comparison, where differences in label granularity cannot be reconciled within a purely categorical framework.

These shortcomings have motivated a growing body of work that seeks to represent emotion in continuous spaces, where distance, direction, and magnitude encode affective relationships. By having a representation methods that preserves intensity, similarity and compositionality, we can enable models to capture nuanced emotional variation beyond rigid class boundaries. This transition from discrete labels to continuous emotion spaces is the focus of the next subsection.

2.1.2 From Discrete Labels to Continuous Emotion Spaces

Continuous emotion representations seek to encode emotion as points in a low-dimensional vector space, where distance, direction, and magnitude correspond to psychological relationships. Early psychological models such as the pleasure–arousal–dominance (PAD) framework [7] formalised emotion along continuous axes, suggesting that emotions differ quantitatively rather than categorically.

1. **Pleasure** represents the hedonic quality of an experience, ranging from unpleasant to pleasant. States such as joy, contentment, and satisfaction occupy the positive end of the pleasure axis, whereas sadness, disgust, and despair lie toward the negative end.

2. **Arousal** captures the degree of physiological and cognitive activation associated with an emotional state, ranging from calm or lethargic to excited or agitated. High-arousal states include fear, anger, and excitement, while low-arousal states include relaxation, boredom, or melancholy.

3. **Dominance** reflects the extent to which an individual feels in control of, or overpowered by, a situation. High-dominance states are associated with feelings of agency, confidence, and control (e.g. pride, anger), whereas low-dominance states correspond to submission, helplessness, or vulnerability (e.g. fear, shame).

Together, these three dimensions define a continuous affective space in which canonical emotion categories correspond to regions rather than points. For example, anger and fear are both negatively valenced and highly aroused, but are separable along the dominance axis; sadness and boredom share low arousal and negative valence, but differ in dominance and intensity. This geometric interpretation naturally accounts for graded emotional intensity, mixed emotions, and smooth transitions between affective states – phenomena that are difficult to represent with discrete labels.

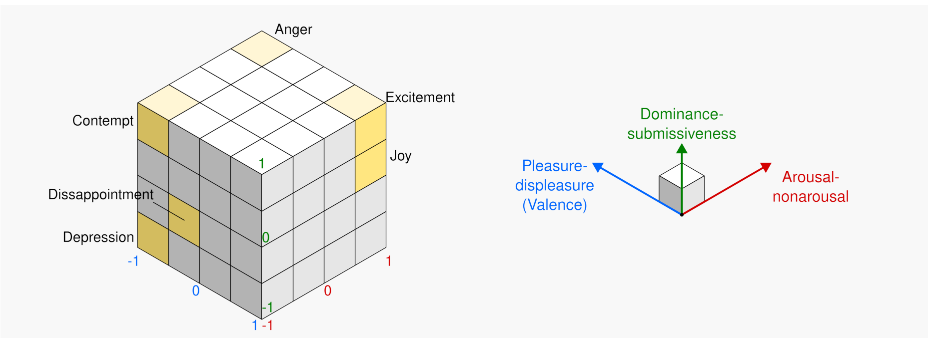

Figure 2.3: Representation of PAD model done with cubes [8]

Additionally, in the field of affective computing, an explicit attempt to bridge the gap between discrete labels and continuous emotion representations is presented by Buechel et al. [9]. Rather than committing to a single emotion taxonomy, they propose learning *label-agnostic emotion embeddings* that serve as a shared latent representation across heterogeneous annotation schemes, including polarity, basic emotion categories, and affective dimensions such as valence and arousal.

Crucially, emotion labels in this framework are not treated as targets to be predicted directly, but as projections from a common embedding space. Models are trained to map text into this latent affective space, from which multiple label formats can be recovered via lightweight prediction heads. This design enables the learned emotion information to be transferred to another without retraining the full model. Empirically, Buechel et al. show that such embeddings preserve prediction quality while supporting cross-dataset generalisation.

Complementary to this, Wang and Zong (2021) [6] focus on learning continuous representations for *emotion categories themselves*. Instead of representing emotion classes as one-hot vectors, they derive distributed emotion embeddings by averaging soft classifier outputs over all instances of a given emotion category. These representations form an emotion space in which distances reflect empirical similarity between emotions as expressed in text. Their results demonstrate that emotion categories cluster by sentiment polarity and intensity, which captures the emotions more faithfully than the conventional semantic word embedding spaces.

Both of those approaches we looked at enable emotion to be represented as a continuous, low-dimensional structure where each of them corresponds to a region or a direction in a metric space. This perspective aligns closely with psychological theories such as PAD, while remaining grounded in data-driven learning from natural language.

Importantly, representing emotion in continuous spaces shifts the learning problem away from rigid classification toward geometric representation learning. This not only resolves many of the limitations inherent in discrete labels, but also provides a foundation for analysing how emotional information is encoded in vector representations more generally. In particular, it raises the question of whether emotion occupies identifiable subspaces or linear directions within learned embeddings, a topic explored in the following subsection.

2.1.3 Linear Structure of Emotion in Word Embedding Spaces

An important question here is whether the continuous space for emotion is actually reflected in learned language representations. Interestingly, prior to the emergence of LLMs, static word embeddings already exhibited evidence of latent affective organisation, despite being trained without explicit emotional supervision.

Wu and Jiang [10] demonstrate that emotional information in word embeddings such as GloVe [11] is not uniformly distributed across dimensions, but instead concentrated along a small number of linear directions. They train a linear autoregressive model to predict continuous emotional valence on annotated narrative data, and show that only a limited subset of embedding dimensions contributes to emotional valence. Interpreting the learned weights reveals that emotional polarity can be captured by a low-dimensional subspace embedded within the full semantic space.

It is also reported that projecting word embeddings into this learned emotion subspace produces a substantially clearer separation between positive and nefative emotion. This indicates that emotion is linearly recoverable even when embeddings are trained without affective objectives. Namely that, emotional meaning is encoded **geometrically**, rather than a byproduct of lexical associations.

A particularly significant finding is that this projected emotion space preserves *linear compositional structure*. Vector arithmetic over words that contain emotion yields composite affective states, which is also consistent with psychological theories of emotion composition such as Plutchik's wheel of emotions [12].

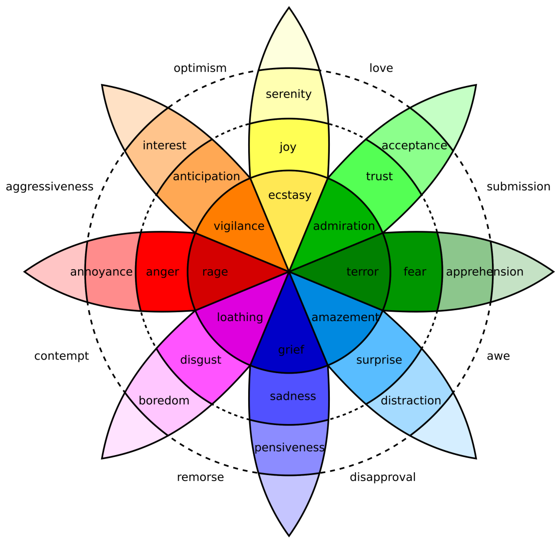

Figure 2.4: Plutchik's wheel of emotions: basic emotions are arranged radially, with angular proximity indicating similarity and radial intensity indicating emotional strength.

For example, the vector sum of `joy` and `trust` aligns closely with `love`, while remaining distant from affectively opposing states such as `remorse`. This behaviour cannot be captured by categorical representations, but emerges naturally in a continuous, vector-valued framework.

These results indicate that emotion in word embedding spaces is not only continuous but also approximately linear, admitting meaningful projections, directions, and arithmetic operations. As a result, affective similarity, intensity, and composition can be expressed through simple linear operations rather than discrete class boundaries.

Taken together, these findings provide early evidence that emotion is encoded as structured geometry within learned language representations. This establishes a representational bridge between psychological models of emotion and mathematical embedding paradigms. This also motivates further research into how such linear emotional structure scales and evolves in more expressive representation learning systems.

2.2 Manipulation of Latent Representations

To harness these emotion vectors, researchers have explored techniques for directly manipulating internal representations of models.

2.2.1 Neuron-Level Emotion Control

One line of work focuses on exploiting localised affective representations at the neuron level. For instance, Radford et al. (2017) [13] noted a single "sentiment neuron" in a byte-level recurrent language model. It has shown that the activation strongly correlates with review polarity and the manipulation of that sentiment neuron directly controls the generated sentiment. Later work shows that modern LLMs contain clusters of neurons selectively responsive to specific emotions.

Ablation and masking experiments were done by Lee et al. (2025) [14], and it demonstrates that removing these neurons can selectively impair recognition of particular emotions. This confirms that they play a functional role rather than being incidental correlates. However, there is still a possibility that the model is redundant. For some emotions, performance remains stable after ablation, suggesting overlapping representations and compensatory mechanisms.

Additionally, low-resource fine-tuning methods such as LoRA (Low-Rank Adaptation) [15] and prefix-tuning operate at a higher level of abstraction but still manipulate latent activations. By injecting small learned vectors into the model we can induce a consistent emotional tone without full retraining. This shows that affective style is encoded in directions that are easy to steer with low-rank updates.

2.2.2 Emergent Emotional Subspaces in Large Language Models

Large language models substantially extend and generalise the affective structure previously observed in static word embeddings (2.1.3). Even though models are trained on massive corpora without any emotional information supervision, LLMs acquire rich contextual representations. These models appear to internalise emotion as a continuous and distributed property of their hidden-state geometry.

Recent work provides strong empirical evidence for this claim. Reichman et al. (2025) [16] show that emotional information in large decoder-only LLMs is organised within a low-dimensional **emotional manifold** embedded in the activation.

They analysed hidden states across layers, demonstrated how affective content aligns with a small number of stable latent directions. Importantly, these directions appeared without any explicit emotion supervision, indicating that affective structure is a natural consequence of large-scale language modelling rather than a task-specific artefact. Moreover, the identified emotional subspace is remarkably consistent across domains and languages, suggesting that LLMs internalise a broadly universal affective geometry.

Crucially, this geometry is not merely descriptive. Reichman et al. [16] show that small interventions along these latent directions can alter the model's emotional interpretation of an input, while preserving its semantic content. This establishes emotional subspaces as functionally meaningful components of the model's representation space, rather than passive correlates of surface-level sentiment cues. This establishes emotional subspaces to be considered as a functional and meaningful components of the model's representation space.

Complementary evidence comes from neuron-level analyses too. Lee et al. (2025) [14] identify groups of neurons whose activations selectively track specific basic emotions in large instruction-tuned models. Emotional information is broadly distributed, but certain neurons constantly exhibit disproportionately strong responses to particular affective states. Further ablation experiments revealed that suppressing these neurons can significantly degrade emotion recognition performance for some emotions. This provides a causal evidence that emotional information is actively processed rather than epiphenomenal.

Taken together, these findings suggest that emotion in LLMs is best characterised as an emergent, low-dimensional subspace embedded within high-dimensional contextual representations. Emotional meaning is not localised to single units, nor reducible to discrete labels; instead, it occupies a structured region of stable latent space. This emergent affective geometry provides the representational foundation that enables direct control of emotional behaviour through linear interventions, which we discuss next.

2.2.3 Steering Latent States with Linear and Continuous Control Vectors

If emotional meaning in LLMs is encoded as directions or subspaces within latent representations, a natural consequence is that affective behaviour should be directly controllable by intervening along those directions. This observation has motivated a line of work on **latent steering**, in which linear and continuous control vectors are applied directly to internal activations during inference.

An early and influential approach is Plug-and-Play Language Models (PPLM) [17], combines a frozen language model with an external attribute model (e.g. a sentiment classifier) and performs gradient ascent on the hidden states to increase the likelihood of a desired attribute. Effectively, this method introduces a small **sentiment vector** into the latent representation, biasing generation toward positive or negative emotion without modifying the model's weights. Although originally framed as a controllable generation technique, PPLM implicitly relies on the assumption that affective attributes correspond to approximately linear directions in latent space.

More recent work makes this assumption explicit and removes the need for iterative optimisation. Anthropic's **Contrastive Activation Addition** (CAA) [18] directly intervenes in a model's internal activations using precomputed **steering vectors**. These vectors are obtained by collecting large sets of positive and negative examples of a target behaviour or attribute and averaging the differences in their intermediate representations. The resulting vector captures a direction in latent space that causally corresponds to the desired attribute. During inference, adding this vector to the residual stream steers model behaviour in a predictable manner, while leaving the model parameters unchanged.

A key advantage of CAA is its simplicity and interpretability. The intervention is linear, and the strength of the effect can be modulated continuously by scaling the steering vector with a scalar coefficient. Moreover, multiple steering vectors can be combined additively, enabling compositional control over several attributes simultaneously. In contrast to prompt engineering or fine-tuning, such interventions operate directly on the model's internal state and often yield more precise and robust control, particularly for abstract properties such as sentiment or emotional tone.

Crucially, steering latent states does not only affect surface-level lexical choices. Injecting affective control vectors at intermediate layers can alter the model's internal interpretation of emotional context, changing how emotional cues are integrated and propagated through the network. This supports the view that emotion in LLMs functions as a global latent variable, rather than as a property tied to individual tokens or outputs.

As conclusion, latent steering methods such as PPLM and CAA provide strong evidence that emotional subspaces in LLMs are not only emergent and structured, but also causally actionable. Manipulating affective behaviour through simple linear interventions reinforces the interpretation of emotion as a continuous geometric representation space. It further bridges descriptive analyses of emotional geometry with concrete mechanisms for controlling model behaviour.

2.2.4 Causal Evidence for Linear Sentiment and Emotion Directions

The existence of affective directions in representation space can be shown by linear probes or dimension reduction, however, it is still doubtful if they actually play a causal role in model behaviour. Recent work provides strong evidence that emotion directions in LLMs are not only linearly recoverable, but also functionally necessary and sufficient for affective reasoning.

Hollinsworth et al. (2024) [19] show that sentiment in LLMs is encoded along a single dominant linear direction in contextual embeddings. They perform directional ablations and interventions, demonstrating that removing / suppressing this sentiment direction causes systematic degradation in downstream emotion-sensitive tasks such as emotion classification. Conversely, injecting the same direction into neutral contexts induces predictable shifts in model outputs. This bidirectional manipulation provides an evidence that the emotion direction is actively used by the model rather than being a passive correlate.

Importantly, affective competence in LLMs does not depend on explicit fine-tuning for emotion-related tasks. Broekens et al. (2023) [20] demonstrate that off-the-shelf

LLMs such as ChatGPT can map text to precise valence–arousal–dominance (VAD) values and detect fine-grained emotional cues using prompting alone. This sugegsts that continuous affective representations are already embedded in the latent space as part of general language understanding. This is a strong evidence that interpretes emotion as a pre-verbal latent variable, rather than a task-specific output format.

At larger scales, evidence further indicates that emotional representations are not flat but hierarchically structured. Zhao et al. (2024) [21] observe that very large models (more than 405 billion parameters) develop tree-like emotion representations, where basic affective states branch into more specific emotions. Models featuring a more consistent hierarchical affective geometry demonstrate enhanced emotion recognition capabilities and increased persuasive effectiveness in negotiation scenarios. This directly links the internal structure of emotional expression to functional behaviour.

These findings converge on a causal interpretation of affective geometry in LLMs. Removing emotion directions impairs affective reasoning, while injecting them predictably alters behaviour. Moreover, emotional information is not confined to isolated tokens or neurons, but is accumulated and summarised across layers into global latent variables that shape general affective capabilities of models.

In conclusion, there is a principled foundation for vector-based manipulation of emotion, which enables us to express emotional nuance through numeric operations on embeddings rather than token-level substitutions. This causal grounding is essential for our project that seeks to control and align emotional behaviour in large language models.

2.3 Reasoning Beyond Token Space

2.3.1 Limitations of Token-Based Reasoning

Conventional LLMs reason in token space. Because inputs are projected onto the discrete boundaries humanity drew as “vocabulary,” fine affective nuance and embodied subtleties are irretrievably lost during that projection, and what remains in one output language may differ from what remains in another. To adjust tone, one often uses prompt engineering or instruction tuning (“Please answer politely,” etc.), but adherence is uncertain and fine-grained control is difficult. Other countermeasures include supervised fine-tuning to internalise a style, or label-conditioned generation (e.g. adding `<joy>` `<sad>`), but fine-tuning is costly and switching between styles is inflexible; label conditioning also cannot capture complex blends of nuance.

Beyond affective control, the same limitation applies to reasoning itself. Autoregressive language models do not take token functionality into account in its generation process. Even if the token carries significant reasoning content, it will be allocated approximately same computational budget as grammartical tokens like `a` or `is`. As a result, only a small subset of internal states carries the actual inferential work, and most of the tokens in explicit reasoning traces do not contribute to the underlying computation.

Recent works show this disconnectivity between the surface output tokens and internal reasoning process. Pfau et al. (2024) [22] demonstrate that transformers can perform non-trivial hidden computation even when output tokens are semantically uninformative. For example, inserting filler or pause tokens (carry no explicit meaning) somehow measurably improve performance on reasoning tasks. This suggests that the essential computation is carried by latent activations rather than by the emitted tokens themselves. Token generation only a communicative language constraint around an internal process.

The inefficiency of token-based reasoning becomes more apparent when reasoning is explicitly verbalised. Chain-of-thought prompting improves performance in many tasks, however **it requires the model to externalise intermediate states into natural language, even when those states are not naturally linguistic**. As argued by Hao et al. (2025) [23], language is optimised for communication rather than for computation. Forcing reasoning to unfold within the constraints of a discrete vocabulary introduces unnecessary overhead and distortion. Many reasoning tokens exist not because they are required for inference, but because they are required for human readability.

From this perspective, explicit verbalisation constitutes a form of lossy compression for an affective reasoning as discussed in the previous section. The continuous and high-dimensional internal states during inference must be projected onto a discrete and culturally contingent token space, which flattens distinctions that are visible for computation but words-level invisible. This compression is problematic for affective computing, because emotion structure is inherently graded, which is resistant to precise lexicalisation.

Taken together, these observations suggest that token space not a substrate of reasoning, but rather an interface. While it provides a convenient medium for communication and supervision, it is suboptimal for internal computation. This motivates a growing line of work that seeks to decouple reasoning from explicit token generation. Computation can proceed without the constraints imposed by language.

2.3.2 Reasoning in Continuous Latent Space

Motivated by the observation that reasoning is primarily carried by latent activations rather than surface tokens, a growing body of work has explored **implicit** forms of chain-of-thought. The central idea is to retain the intermediate reasoning within the model's hidden state, so that we can benefit from the multi-step reasoning while avoiding the distortions introduced by forcing intermediate states into natural language.

Early implicit chain-of-thought approaches aim to internalise reasoning traces by gradually removing explicit reasoning tokens during training, encouraging the model to perform the same computation silently in its latent space. More recent methods go further by explicitly allocating computation to continuous latent representations that replace textual reasoning steps altogether. In these models, reasoning unfolds as a sequence of hidden-state transitions, while the output layer is used only to produce the final answer.

A representative work is **CODI** (Compressing Chain-of-Thought into Continuous Space via Self-Distillation) [24]. This approach compresses explicit chain-of-thought trajectories into a compact sequence of continuous latent states, and by distilling a teacher model that produces explicit reasoning traces, CODI trains a student model to reproduce the same reasoning outcome using a shorter sequence of latent vectors. This demonstrates that the information carried by textual reasoning can be encoded more efficiently in continuous space, without requiring token-level supervision at inference time.

A more radical approach is proposed by **Coconut** (Chain of Continuous Thought) [23]. Rather than compressing explicit reasoning traces, Coconut modifies the inference procedure itself by feeding the model's last hidden state back as the next input embedding. In this way, reasoning unfolds entirely within continuous latent space and there is no need of intermediate textual tokens at all. However, because Coconut relies primarily on answer-level supervision, the resulting latent trajectories can be unstable when extended to deeper reasoning chains, which we discuss in the next subsection.

2.3.3 Implicit and Continuous Chain-of-Thought

However, there is one challenge that purely implicit reasoning introduces. When multiple latent reasoning steps are chained together without sufficient structural constraints, the latent representations can collapse into homogeneous states. This phenomenon is often referred to as **latent collapse**, which undermines the model's ability to maintain distinct intermediate representations.

SIM-CoT (Supervised Implicit Chain-of-Thought) [25] addresses this limitation by introducing step-level supervision during training. Instead of supervising only the final answer or the overall reasoning trajectory, SIM-CoT attaches an auxiliary decoder that forces each latent reasoning step to predict a corresponding explicit reasoning phrase. This alignment ensures that each latent vector encodes distinct, semantically meaningful content, preventing homogenisation and improving stability. Crucially, the auxiliary decoder is used only during training and is removed at inference, allowing the model to retain the efficiency of silent latent reasoning while benefiting from structured supervision. Empirically, SIM-CoT substantially improves both performance and robustness over earlier latent reasoning models, including Coconut-style architectures.

It attaches an auxiliary decoder that forces each latent reasoning step to predict a corresponding explicit reasoning phrase, which ensures that each latent vector to encoded semantically meaningful content. This substantially improves both performance and robustness over earlier latent reasoning models, including Coconut-style architectures. Importantly, the auxiliary decoder is used only during training and is removed at inference, allowing the model to retain the efficiency of silent latent reasoning while benefiting from structured supervision.

These approaches demonstrate that chain-of-thought need not be realised as a sequence of discrete tokens. Reasoning can be instead implemented as a controlled trajectory through continuous latent space, with explicit language serving primarily as a training scaffold or interpretability aid. Implicit and continuous chain-of-thought models provide a natural foundation for studying how emotions may be represented and manipulated within latent reasoning processes.

2.3.4 Implications for Affective Representation

The developments reviewed above suggest a converging picture of representation and computation in LLMs. As discussed in earlier sections, emotions are not confined to discrete labels or lexical choices but it rather emerges as a continuous structure within latent space. At the same time, recent advances in implicit chain-of-thought demonstrate that reasoning itself can be implemented as a trajectory through that same latent space, without requiring intermediate symbolic tokens.

These findings summarises to the fact that emotion need not be treated as separate components at the level of surface language. Instead, it appears to be grounded in shared latent representations that evolve over the course of inference. From this perspective, emotional information may constitute an integral part of the internal reasoning state itself, shaping how intermediate representations are formed and propagated.

Importantly, this possibility remains largely unexplored. Although latent reasoning frameworks have demonstrated improved eff expressivity for logical and mathematical tasks, their implications for affective representation have not been systematically examined. Whether emotional dimensions align or interact with latent reasoning trajectories is an open question. Addressing this gap requires moving beyond token-level analysis and considering emotion as a first-class component of latent computation rather than as a stylistic attribute.

These considerations motivate the investigation in the following chapters. Rather than treating emotion as an external label or an output constraint, we explore the hypothesis that affective states can be represented within continuous latent spaces that underlie reasoning itself. By doing so, we aim to better understand how emotional meaning may be re-expressed when language is no longer the primary medium of inference.

Chapter 3

Affective Steering Pipeline

The purpose of this chapter is to demonstrate that we can construct an emotional direction within the residual space of a language model, and use it to actually steer the output. We do not delve deeply into the semantic interpretation of the emotion representations themselves yet. Rather, our empirical claim is that, for a given target emotion e, applying a CAA direction derived from contrastive residual differences can shift the affective framing of the output towards the target emotion while largely preserving semantic content.

The central conclusion of this chapter is that by decomposing emotion-specific CAA vectors into a shared emotionalisation component and an emotion-specific residual component, we can bias the output toward a target emotion. Namely that,

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G g + \alpha_R \hat{r}_e$$

where g is the shared emotionalisation vector that shifts from neutral to emotional expression, and $\hat{\mathbf{r}}_e$ is the residual vector specific to the target emotion. In this chapter, we treat this decomposition as a practical and operational representation for steering, rather than making strong claims about the underlying semantics of the emotion representations themselves.

Experimental Roadmap

This chapter is organised as follows.

1. **Dataset construction**: From a single base utterance, we generate emotional rewrites corresponding to multiple target emotions and intensity levels, and further construct neutral paraphrases. This allows us to create a controlled corpus that enables the comparison of emotional variations while preserving the original meaning.

2. **Residual-stream extraction**: We combine an emotion rewrite and the corresponding neutral paraphrase into a prompt of the same format and extract residual-stream activations from selected layers of Llama-3.1-8B-Instruct. For each source utterance, we compute

$$\mathbf{H}^{\mathrm{emo}}_{n,e,i,l} - \mathbf{H}^{\mathrm{neu}}_{n,l}$$

to obtain an affective shift that cancels out the semantic content as much as possible.

3. **Layer selection and diagnostics**: We compare diagnostic metrics such as emotion-category probe accuracy and centroid compactness, to select the primary analysis layer. The goal here is not to determine the most interpretable layer, but rather to determine the layer that provides the most stable affective direction for steering.

4. **CAA vector construction**: We construct contrastive activation addition (CAA) vectors by averaging the residual vectors over sources and intensities, and then normalising to obtain unit vectors for each emotion.

5. **Operational decomposition for steering**: Since cross-emotion CAA directions exhibit very high cosine similarity, this suggests that each emotion direction contains a large shared emotionalisation component. Therefore, we define

$$\mathbf{g} = \frac{\frac{1}{|E|}\sum_{e\in E}\mathbf{c}_e}{\left\|\frac{1}{|E|}\sum_{e\in E}\mathbf{c}_e\right\|_2}$$

and remove this shared component from each emotion direction.

$$\hat{\mathbf{r}}_e = \mathbf{c}_e - (\mathbf{c}_e^\top \mathbf{g})\mathbf{g}$$

In this chapter, we operationally use this $\hat{\mathbf{r}}_e$ as the **steering residual** responsible for the target emotion identity. However, the internal affective structure of $\hat{\mathbf{r}}_e$ will be discussed in the next chapter.

6. **Steering experiments**: We add the steering vector $\alpha_G\mathbf{g} + \alpha_R\hat{\mathbf{r}}_e$ to the residual stream and evaluate whether the generated output shifts toward the target emotion.

3.1 Dataset Construction

In order to investigate whether emotion is encoded as structured geometry in residual space, we constructed controlled rewrite datasets from shared source utterances. The primary dataset is an emotion-category rewrite set derived from two dialogue corpora: DailyDialog [26] and EmpatheticDialogues [27]. We selected dialogue data because conversational utterances provide compact emotional signals grounded in context in everyday interpersonal situations, while keeping propositional content sufficiently constrained for controlled rewriting. For each source utterance, we used GPT-4.1-mini to generate rewrites for Plutchik's eight basic emotions (joy, trust, fear, surprise, sadness, disgust, anger and anticipation) at three matched intensity levels (low, medium and high). This yields $8 \times 3 = 24$ affective variants per source text. A central constraint throughout the construction process was to preserve meaning: the underlying situation and factual content were held constant while only the affective framing was systematically varied.

Emotion	**Utterance**
Base	**I waited for hours and nobody came.**
Joy	Nobody came for hours, but I ended up enjoying the quiet time on my own.
Trust	Nobody came for hours, but I am sure they had a valid reason.
Fear	Nobody came for hours, and I started worrying that something might have gone wrong.
Surprise	Nobody came for hours, and honestly I did not expect that at all.
Sadness	For hours, nobody came. I guess that says enough.
Disgust	Nobody came for hours; that felt deeply inconsiderate.
Anger	Nobody came for hours, and I am really frustrated about wasting all that time.
Anticipation	Nobody came for hours, but I kept thinking they might still arrive any minute.

Table 3.1: Illustrative rewrites for all eight Plutchik emotion categories at matched medium intensity.

As a control condition, we constructed an additional neutral paraphrase baseline dataset. In this dataset, each source utterance has been rewritten into an affectively neutral but semantically equivalent form. This baseline provides a reference representation of content without targeted emotional colouring, enabling contrastive analysis between emotion-conditioned and emotion-suppressed activations.

Field	**Example**
Base utterance	I waited for hours and nobody came.
Neutral paraphrase	I waited for several hours, and no one arrived.
Purpose	Preserve events and meaning while removing affective phrasing.

Table 3.2: Illustrative example from the neutral-paraphrase control dataset.

Together, these datasets enable us to distinguish emotional signals from lexical-semantic content. This provides an empirical basis for the decodability, geometry and steering analyses presented in subsequent sections.

This design makes it possible to ask whether affective variation forms a consistent direction in activation space, rather than merely reflecting surface-level lexical differences between emotional words.

3.2 Residual-Stream Extraction

We extract hidden-state activations from an instruction-tuned language model (Llama-3.1-8B [28]) with no gradient updates. Our aim is to determine where affective information is represented in the model's internal computations rather than in output tokens alone. The residual stream is therefore used as the primary analysis target, because it is the shared channel through which each transformer block accumulates and propagates semantic and affective information across layers.

To ensure that the inputs are comparable across the different conditions, each sample is formatted with a shared instruction prefix that is constructed from the same base utterance. Only the continuation text is changed, either through an emotion rewrite or a neutral paraphrase. Specifically, we use a user instruction in the form of a chat template that asks for a rewrite aligned with the speaker's feelings, and then append the rewritten sentence as the assistant continuation. This matching protocol controls for variance at the prompt level and enables direct comparison of emotion-conditioned and neutral-conditioned activations.

Template Prompt
Please rewrite this so it aligns with my feelings more. Start straight from the rewritten text without any preamble, and only provide **one** rewritten version. Base: $\{\texttt{base\_text}\}$

Table 3.3: Prompt format used for activation extraction.

Residual sampling involves forwarding batched inputs with hidden-state output enabled and collecting vectors at selected hook layers. For each target layer l, the last token's hidden state is taken from the layer's output, i.e.

$$\mathbf{h}_l = \texttt{hidden\_states}[l+1][:, -1, :]$$

Since tokenisation uses left padding, the index -1 consistently corresponds to the final real token in each sequence in the batch. Final-token representation is the state after the model has processed the entire rewritten utterance, making it a compact summary of the affective framing expressed by the continuation. Activations are sampled from layers $\{8, 10, 13, 16, 19, 22\}$, producing a multi-layer representation for every input.

The emotion-intensity extraction pipeline produces a 5D tensor:

$$\mathbf{H}^{\text{emo}} \in \mathbb{R}^{N \times E \times I \times L \times D},$$

where N is the number of source texts, $E = 8$ emotions, $I = 3$ intensity levels, L is the number of sampled layers, and D is the hidden dimension. The neutral baseline pipeline produces:

$$\mathbf{H}^{\text{neu}} \in \mathbb{R}^{N \times L \times D}.$$

Source IDs are sorted to align both tensors by index, allowing direct subtraction

$$\Delta\mathbf{H}_{n,e,i,l,:} = \mathbf{H}^{\text{emo}}_{n,e,i,l,:} - \mathbf{H}^{\text{neu}}_{n,l,:},$$

which isolates affective shifts while controlling for base semantic content.

3.3 Layer Selection and Diagnostics

Not all transformer layers provide equally useful representations for affective analysis. Emotion-related structure may be present at multiple depths, but different layers can privilege different properties of the representation. For example, one layer may provide strong emotion-category separability, while another may better preserve a compact low-dimensional geometry. Since the later analyses require both linear decodability and geometrically stable affective directions, we compared candidate layers using several complementary diagnostics.

Based on these diagnostics, layer 13 was selected as the primary layer for the main analyses. It is not globally best on every metric, but it provides the most convincing overall trade-off between category separability, low-dimensional compactness, and sensitivity to graded affective variation.

Layer	**Emotion probe BAcc** *(4-D subspace)*	**Centroid EVR$_4$**	**Mean intensity-direction correlation**
10	0.6751	0.8876	0.0998
13 *	**0.7033**	0.8920	**0.1336**
16	0.6704	**0.8958**	0.1297

Table 3.4: Layer-wise diagnostic summary for candidate layers. Emotion-category probe balanced accuracy (4-D subspace), centroid compactness (EVR$_4$), and mean local intensity-direction correlation. Bold indicates the best value per column; asterisk marks the selected reporting layer.

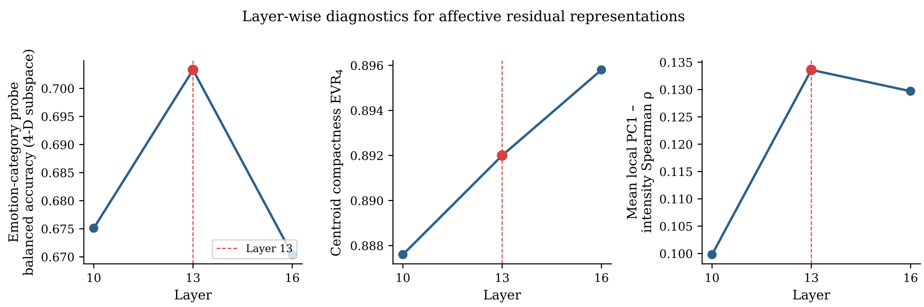

Figure 3.1: A visualisation of the layer diagnostics across candidate layers.

First, emotion-category probe performance in the 4D affective subspace is highest at layer 13. The balanced accuracy reaches 0.7033 at layer 13, compared with 0.6751 at layer 10 and 0.6704 at layer 16. On this criterion, layer 13 preserves the most linearly decodable information about the target emotion categories.

Second, this increase in decodability does not come at the cost of geometric compactness. The centroid-level four-dimensional explained variance remains high at layer 13, with $\mathrm{EVR}_4 = 0.8920$. This is slightly below layer 16 ($\mathrm{EVR}_4 = 0.8958$) but above layer 10 ($\mathrm{EVR}_4 = 0.8876$). In other words, layer 13 is not the strongest layer on this metric alone, but it remains close to the best-performing depth while retaining stronger emotion-category separability.

Third, layer 13 shows the strongest sensitivity to graded affective variation among the candidate layers. The mean intensity-direction correlation is 0.1336 at layer 13, compared with 0.0998 at layer 10 and 0.1297 at layer 16. Although the absolute correlation is modest, the relative increase suggests that layer 13 is the most sensitive of the three candidate layers to systematic changes in affective intensity.

To avoid overfitting the analysis to a single depth, we also examined neighbouring layers as robustness checks. Layers 10 and 16 show similar qualitative trends and preserve the main geometric findings, suggesting that affective structure is not confined to a single transformer block. We therefore do not treat layer 13 as uniquely optimal in a global sense. Rather, we use it as the main reporting layer because it offers the best trade-off across the three diagnostics, while layers 10 and 16 serve as nearby robustness checks.

3.4 CAA Vector Construction

In order to convert contrastive residual differences into controllable steering signals, we derive Contrastive Activation Addition (CAA) directions from emotional and neutral conditioned activations. For each source utterance n, emotion e, intensity level i, and layer l, we first define

$$\mathbf{d}_{n,e,i,l} = \mathbf{h}^{\mathrm{emo}}_{n,e,i,l} - \mathbf{h}^{\mathrm{neu}}_{n,l}.$$

We then average over sources and normalise to unit length:

$$\bar{\mathbf{d}}_{e,i,l} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{d}_{n,e,i,l}, \qquad \mathbf{c}_{e,i,l} = \frac{\bar{\mathbf{d}}_{e,i,l}}{\|\bar{\mathbf{d}}_{e,i,l}\|_2}.$$

The vectors $\mathbf{c}_{e,i,l}$ are intensity-specific CAA directions.

As the steering stage primarily targets the emotional category, treating intensity as a secondary modulation, we also build pooled directions by averaging across intensities and re-normalising:

$$\mathbf{c}^{\mathrm{pool}}_{e,l} = \frac{\frac{1}{I}\sum_{i=1}^{I} \bar{\mathbf{d}}_{e,i,l}}{\left\|\frac{1}{I}\sum_{i=1}^{I} \bar{\mathbf{d}}_{e,i,l}\right\|_2}.$$

In the main steering analyses, these pooled unit vectors are used as the primary candidate steering vectors.

Empirically, this construction remains stable at the selected reporting depth (layer 13). Alignment between local contrastive shifts and global CAA directions is high for all emotions at layer 13, with mean cosine values ranging from 0.8536 to 0.8570 and standard deviations of around 0.054. This is slightly stronger than layer 10 and much stronger than deeper candidate layers, such as layer 16 and above. This matches the result of the layer selection that layer 13 provides the best overall trade-off for controllable affective structure.

The pooled directions are also highly consistent across intensity levels. At layer 13, the cosine similarity between low, medium, and high CAA vectors remains very high for all emotions (approximately 0.983–0.999 across all intensity pairs). This suggests that intensity changes mainly in terms of magnitude around a shared category direction, rather than substantially altering the direction itself.

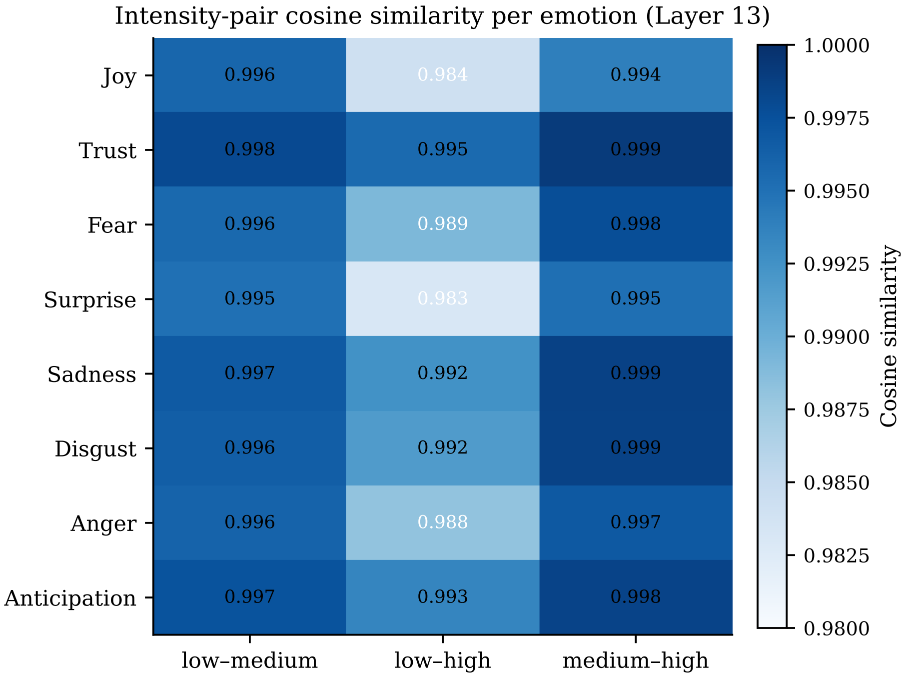

Figure 3.2: CAA intensity-pair cosine similarity per emotion at layer 13

Intensity pair	Mean cosine	Min cosine	Max cosine
low-medium	0.996	0.995	0.998
low-high	0.990	0.983	0.995
medium-high	0.997	0.994	0.999

Table 3.5: Summary of intensity-pair cosine similarity at Layer 13, computed across all eight emotion categories. Even for the largest intensity gap (*low-high*), direction consistency remains above 0.98, indicating that intensity modulation primarily scales rather than redirects the CAA vector.

Finally, the cross-emotion cosine similarities at layer 13 are also high, ranging from approximately 0.979 to 0.998. This indicates that pooled CAA directions share a dominant common component across categories. We therefore use $\mathbf{c}_{e,13}^{\text{pool}}$ as practical steering candidates and explicitly decompose them in the next section.

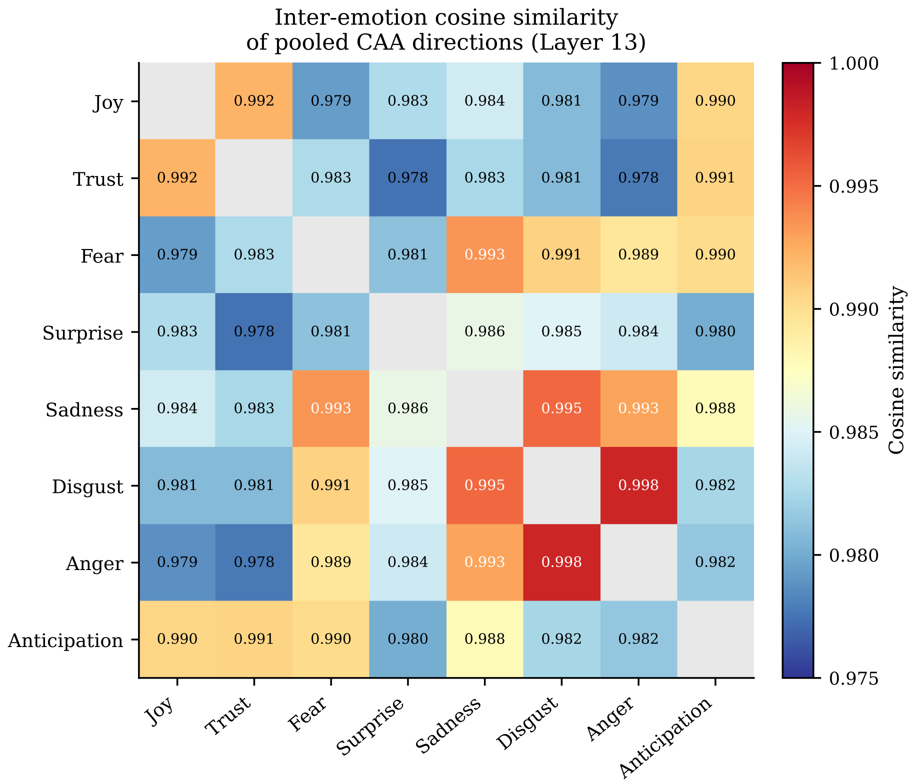

Figure 3.3: Inter-emotion cosine similarity of pooled CAA directions at layer 13

3.5 Operational Decomposition for Steering

The pooled CAA directions at layer 13 are highly aligned across emotion categories, suggesting that each direction contains a large shared component corresponding to generic emotionalisation. To separate this common shift from emotion identity, we define a shared direction as the normalised mean over emotion-specific pooled CAA vectors:

$$\mathbf{g} = \frac{\frac{1}{|E|}\sum_{e\in E} \mathbf{c}_{e,13}^{\text{pool}}}{\left\|\frac{1}{|E|}\sum_{e\in E} \mathbf{c}_{e,13}^{\text{pool}}\right\|_2}.$$

We then remove this shared component from each pooled direction by orthogonal projection:

$$\mathbf{r}_e = \mathbf{c}_{e,13}^{\text{pool}} - \left(\left(\mathbf{c}_{e,13}^{\text{pool}}\right)^{\top}\mathbf{g}\right)\mathbf{g}.$$

For steering, we use the unit-normalised residual

$$\hat{\mathbf{r}}_e = \frac{\mathbf{r}_e}{\|\mathbf{r}_e\|_2},$$

and construct interventions as

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e.$$

In this chapter, we treat this decomposition as an operational parameterisation for controllable steering: g controls global emotionalisation strength, while $\hat{\mathbf{r}}_e$ controls emotion-specific directionality. A deeper geometric interpretation of these residual components is deferred to the next chapter.

3.6 Steering Experiments

Generation Setup

This experiment tests whether recovered CAA directions can influence text generation causally. We use a pre-trained Llama-3.1-8B-Instruct model to generate continuations based on manually selected seed utterances that are each mildly biased towards one Plutchik emotion. At this stage, our goal is empirical: to verify that the intervention can reliably bias generations toward target emotions while preserving semantic content, and to quantify how strongly the effect depends on the initial seed context.

To evaluate steering under different initial affective states, we used eight manually selected seed utterances that carry mild initial emotional colouring. For each seed utterance and target emotion, we generate rewritten outputs at various coefficient settings. During inference, we intervene in the residual stream at layer 13 by adding a decomposed steering vector. Generation uses deterministic greedy decoding to ensure reproducibility. To ensure that the model computes all subsequent logits under the modified representation, we intervene at the last token position in the residual stream before generation begins.

This design makes the experiment more challenging and informative because the intervention is evaluated under diverse initial affective states. Therefore, variation in the seed emotion functions as a robustness test for affective steering, rather than merely providing generation prompts.

Intervention Families

The intervention vector for target emotion e and coefficients (α_G, α_R) is

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e,$$

where g is the shared emotionalisation direction and $\hat{\mathbf{r}}_e$ is the unit-normalised emotion-specific residual. We evaluated five intervention families (Table 3.6): none (baseline), shared-only ($\alpha_G \mathbf{g}$), residual-only ($\alpha_R \hat{\mathbf{r}}_e$), original pooled CAA ($\alpha \mathbf{c}_e^{\text{pool}}$), and decomposed ($\alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$). These families enable us to isolate the causal contributions of shared emotionalisation versus emotion-specific residuals.

These families define the condition-wise comparison used in the aggregate analysis, whereas the 1280-sample count refers to the coefficient-sweep generation run.

Intervention Family	**Vector applied to residual stream**
None (baseline)	No intervention; standard generation
Shared-only	$\Delta = \alpha_G \mathbf{g}$ (emotionalisation control, no emotion identity)
Residual-only	$\Delta = \alpha_R \hat{\mathbf{r}}_e$ (emotion identity control, no shared emotionalisation)
Original CAA	$\Delta = \alpha \mathbf{c}_e^{\mathrm{pool}}$ (monolithic pooled direction)
Decomposed	$\Delta = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$ (combined shared + residual control)

Table 3.6: Five intervention families were tested during the generation process. The decomposed family isolates the contributions of shared emotionalisation from emotion-specific residuals. In contrast, the monolithic original CAA family applies the full pooled direction as a single coefficient.

Coefficient Sweeps

For the coefficient-sweep experiment, we evaluated ten unique coefficient pairs (α_G, α_R), constructed from three sweep settings: a fixed-α_G sweep over α_R, a fixed-α_R sweep over α_G, and a small two-dimensional grid over both coefficients. For each coefficient pair, we generated outputs for all 8 seed emotions and all eight target emotions under two intervention application modes. This gives

$$8 \times 8 \times 2 \times 10 = 1280$$

generated samples in total. The five intervention families reported below are used as comparison groups in the aggregate evaluation.

Evaluation Metrics

The core diagnostics are directionality (i.e. target-emotion shift), approximate monotonicity under coefficient change and semantic preservation under intervention. The target-emotion match is evaluated using a fine-tuned emotion classifier, which is trained on the original dialogue corpora and applied to the generated continuations afterwards. Semantic preservation is assessed via semantic similarity (frozen sentence encoder) between the generated text and the seed utterance. Monotonicity is computed as the proportion of coefficient sweeps showing a monotonic increase or decrease in the target-match score as the coefficients increase. We also flag outputs that are empty or severely degraded as failure cases.

Results

Table 3.7 summarises the main steering diagnostics. Overall, steering produces a small but consistent improvement in the match between the target emotion and the actual emotion displayed. The decomposed intervention achieves the highest aggregate target match (0.364), compared with 0.325 for no steering, 0.349 for shared-only steering, 0.357 for residual-only steering, and 0.359 for the original pooled CAA direction. This suggests that the shared emotionalisation direction and the emotion-specific residual have complementary causal effects.

Diagnostic	**Result**
Target-emotion match	No steering: 0.325; shared-only: 0.349; residual-only: 0.357; original CAA: 0.359; decomposed: 0.364
Best aggregate intervention	Decomposed steering, with a $+0.039$ absolute gain over the no-steering baseline
Representative emotion-level gains	fear: $0.220 \to 0.310$; trust: $0.300 \to 0.385$; sadness: $0.560 \to 0.610$
Monotonicity under coefficient scaling	Layer 13 monotonic fraction: 0.6856; nearby checks: 0.7014 at layer 10 and 0.6343 at layer 16
Semantic preservation	No steering: 0.946; steering conditions remain high, within 0.954–0.973
Emotionality	Scores remain within 0.819–0.854, suggesting that steering does not rely on severe text degradation
Robustness	Canonical slice $(\alpha_G = 3, \alpha_R = 3)$ produces no empty outputs across target emotions; cross-seed length variability remains non-trivial, with some targets showing standard deviations of approximately 9–10 words

Table 3.7: Summary of steering results. Decomposed CAA steering yields the strongest target-emotion shift while preserving semantic content. Monotonicity is present but partial, and robustness checks indicate that outputs remain constrained by the seed context.

The monotonicity and robustness results suggest that although coefficient scaling provides some control, it is not entirely effective. Increasing steering strength tends to move generations in the intended emotional direction, but this effect is not consistent across all seeds and targets. Similarly, robustness tests across seed emotions demonstrate that steering does not overwrite the initial affective context. Rather, the intervention biases the generation towards the target emotion, while remaining constrained by the semantic and emotional affordances of the seed utterance.

Qualitative Examples

Table 3.8 shows representative examples from the canonical steering slice ($\alpha_G = 3, \alpha_R = 3$, last-token intervention). These examples illustrate that the intervention does not simply overwrite the source context. Instead, it tends to bend the existing situation toward the target affective direction.

Seed	Target	Seed utterance	Steered output
fear	joy	Walking alone through the dark alley I felt a cold shiver run down my spine.	As I strolled through the dimly lit alley, the flickering streetlights cast a warm glow, but the sense of solitude and tranquility that filled the air made me feel a gentle, soothing calm wash over me.
disgust	joy	The smell from the bin was overwhelming and the sight of it made my stomach turn.	The putrid aroma wafting from the trash can was a welcome respite, and the vibrant colors of the discarded items made my heart feel uplifted.

Table 3.8: Representative examples of decomposed CAA steering in the canonical setting. The fear-to-joy example shows a natural affective shift while preserving the dark-alley context. The disgust-to-joy example shows a stronger and less natural reinterpretation, illustrating both the causal effect and the contextual limits of the intervention.

These examples support that decomposed CAA steering can guide the output towards the target emotion without replacing the original context. For example, in the conversion from fear to joy, the alleyway context is semantically preserved while the emotional framework shifts towards calmness. In the example of converting from disgust to joy, the model attempts a more extreme positive reinterpretation of a disgust-inducing scene, demonstrating directional influence and contextual limitations. This pattern aligns with the quantitative robustness results. In other words, while steering functions effectively at the behavioural level, the final output is still limited by the semantic and emotional properties of the initial utterance.

Interpretation

The results suggest that decomposed CAA steering is a modest yet reliable method of influencing emotional framing. As summarised in Table 3.7, the intervention neither overwrites the seed utterance nor imposes an absolute target emotion template. Instead, it shifts the emotional tone of the existing content while remaining constrained by the semantic and emotional possibilities of the input.

This distinction is important for designing affective rewriting systems. The shared component g appears to control the overall degree of emotionalisation, and the residual component $\hat{r}_e$ contributes target-specific affective directionality. The decomposed components consistently yielded slightly better results than the individual components alone or the original integrated CAA vector. This suggests that emotional control is enhanced by separating global emotional intensity from emotion-specific bias.

At the same time, as reported in Table 3.7 (in particular the monotonicity and robustness rows), the partial monotonicity and seed-dependent variation suggest that the current intervention does not provide complete control. Therefore, future systems should treat steering coefficients as adaptive parameters that can be adjusted according to the initial emotional state of the input rather than applying them uniformly in all situations.

3.7 Limitations and Transition

These results show that affective steering can be implemented as a decomposed residual-space intervention. The shared direction g controls the degree of emotionalisation, while the residual direction $\hat{\mathbf{r}}_e$ provides target-specific affective bias. Together, $\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$ offers a practical mechanism for moving output text toward a target emotional framing, while largely preserving the original semantic content.

However, the results also reveal several limitations. Firstly, the steering effect is modest rather than absolute. While target-emotion matching improves, the model does not reliably converge to the target emotion across all seeds. Secondly, monotonicity under coefficient scaling is only partial, suggesting that larger coefficients do not always produce proportionally stronger affective shifts. Thirdly, the intervention remains strongly dependent on the initial utterance, meaning that affective steering should be considered a contextual bias rather than a complete replacement of the initial emotional state.

These limitations motivate the next chapter. So far, $\hat{\mathbf{r}}_e$ has been treated as a single entity. The next chapter asks what this residual actually represents. In particular, it examines whether emotions such as joy, sadness, and anger should be treated as atomic directions, or whether they contain local sub-axes that correspond to more subtle, pre-verbal affective vectors.

Chapter 4

Pre-Verbal Affective Vector Projection

The previous chapter established that, for a given target emotion e, the decomposed steering vector

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$$

reliably shifts the affective framing of generated text towards e while preserving semantic content. In that framework, joy, sadness, anger, and the other Plutchik categories were treated as **primitive** units. This means that each $\hat{\mathbf{r}}_e$ was taken to be an atomic emotion-specific vector without further decomposition.

This chapter re-examines that assumption. The central question is whether named emotion categories, such as **joy** or **fear**, are:

- the minimal units of affective representation in the model's residual stream, or
- whether each $\hat{\mathbf{r}}_e$ is itself a composite that consists of a small set of more fundamental, pre-categorical affective dimensions.

The experiments reported here provide evidence for the latter. Namely, within each named emotion, there exist local sub-axes, and these sub-axes are largely shared across emotion categories. They can be interpreted as vectors that are more primitive than any named emotion label.

The central conclusion is that what Chapter 3 called $\hat{\mathbf{r}}_e$ can be re-read as a projection onto a shared pre-verbal affective space:

$$\hat{\mathbf{r}}_e \;=\; \sum_{k=1}^{K} \beta_{e,k}\, \mathbf{m}_k \;+\; \boldsymbol{\varepsilon}_e,$$

where $\mathbf{m}_k$ are label-free meta-axes shared across all emotions, $\beta_{e,k} = \hat{\mathbf{r}}_e^\top \mathbf{m}_k$ are scalar projections that encode the "affective fingerprint" of emotion e, and $\boldsymbol{\varepsilon}_e$ is the residual component that is orthogonal to all $\mathbf{m}_k$.

Experimental Roadmap

This chapter is organised as follows.

1. **Geometry of the emotion-specific residual**: We first characterise the geometry of the residual emotion directions after removing the shared emotionalisation component $\mathbf{g}$. At the centroid level, we test whether the residual directions recover a structured Plutchik-like organisation. At the per-source level, we verify that individual $\mathbf{g}$-residualised shifts still carry linearly decodable emotion-label information. Together, these analyses establish that $\hat{\mathbf{r}}_e$ is a genuine emotion-specific signal rather than a residual artefact or noise.

2. **Internal variance structure**: We characterise the internal dimensionality of each named emotion by applying PCA to the distribution of per-source residual vectors within each emotion category.

3. **Local sub-axis extraction and interpretation**: We extract the leading principal directions within each emotion and submit them to an *LLM as a judge* [29] for semantic interpretation. We also confirm that these axes are affective rather than stylistic artefacts.

4. **Cross-emotion sharing**: We show that the local PCA axes cluster by their **rank** rather than by emotion label. This leads us to a set of shared meta-axes $\mathbf{m}_k$ that span all named emotions.

5. **Extraction and formalisation of meta-axes**: We formalise $\mathbf{m}_k$ as sign-aligned means over per-emotion PCA directions, and assess their stability.

6. **Pre-verbal interpretation**: We interpret each $\mathbf{m}_k$ by presenting extreme examples from all emotions simultaneously to an LLM, obtaining emotion independent semantic labels for each meta-axis.

7. **Re-decomposition of $\hat{\mathbf{r}}_e$**: We project each $\hat{\mathbf{r}}_e$ onto the $\mathbf{m}_k$ basis, quantify the explained variance, and state the revised decomposition that extends the formula from Chapter 3.

8. **Causal validation by local axis steering**: We test whether adding $\pm\beta\,\mathbf{u}_{e,1}$ or $\pm\beta\,\mathbf{m}_1$ to the steering vector causally shifts outputs along the expected affective sub-dimension.

9. **Label-free steering**: We compare three steering conditions (named emotion prototype only, $\mathbf{m}_k$-projection only, and their combination) to determine whether the $\mathbf{m}_k$ decomposition retains the practical utility of $\hat{\mathbf{r}}_e$.

4.1 Geometry of the Emotion-Specific Residual

Before analysing the internal sub-structure of $\hat{\mathbf{r}}_e$, we first characterise what kind of signal remains after removing the shared emotionalisation component. This section therefore asks two complementary questions. First, do the eight emotion directions form a structured geometry after residualisation? Second, is this structure supported by individual per-source residual shifts, rather than being only an artefact of averaging?

Raw directions are near-collinear

At layer 13, the cosine similarities between the eight pooled CAA vectors $\mathbf{c}_e^{\mathrm{pool}}$ are extremely high. The mean off-diagonal cosine is 0.986, and even Plutchik opposite pairs, such as joy–sadness and trust–disgust, have a mean cosine of 0.984. This is almost indistinguishable from that of adjacent pairs (0.988). This tendency for the vectors to align almost perfectly in a straight line is characteristic of g, the dominant common factor. In other words, all eight vectors point almost exactly in the same direction because the largest component of emotional activation is simply the transition from "neutral" to "emotional".

Layer	**PR**	**EVR$_{\mathrm{PC1}}$**	**EVR$_{\mathrm{PC1\text{-}3}}$**
10	3.88	0.407	0.797
13 *	**3.61**	**0.437**	**0.812**
16	3.33	0.475	0.825

Table 4.1: Low-dimensionality of the pooled CAA geometry across the main candidate layers. The raw emotion directions consistently occupy a compact subspace rather than eight independent axes.

Despite this near-collinearity, PCA over the eight pooled directions reveals a low-dimensional but non-trivial structure. At layer 13, the participation ratio is 3.61, with $\mathrm{EVR}_{\mathrm{PC1}} = 0.437$ and $\mathrm{EVR}_{\mathrm{PC1\text{-}3}} = 0.812$. Even before performing an explicit decomposition, the emotion directions are therefore not distributed across eight independent dimensions, but are organised within roughly three to four effective axes.

Residualisation exposes structured emotion-specific geometry

After removing g, the geometry changes qualitatively. The residual directions are defined as

$$\mathbf{r}_e = \mathbf{c}_e - (\mathbf{c}_e^{\top}\mathbf{g})\mathbf{g}.$$

These residual directions have a mean off-diagonal cosine of -0.137. Plutchik opposite pairs become clearly negative on average (-0.306), while adjacent pairs remain weakly positive ($+0.090$). This reversal of sign shows that the residual space encodes an emotion-specific structure consistent with Plutchik's theoretical organisation of emotions.

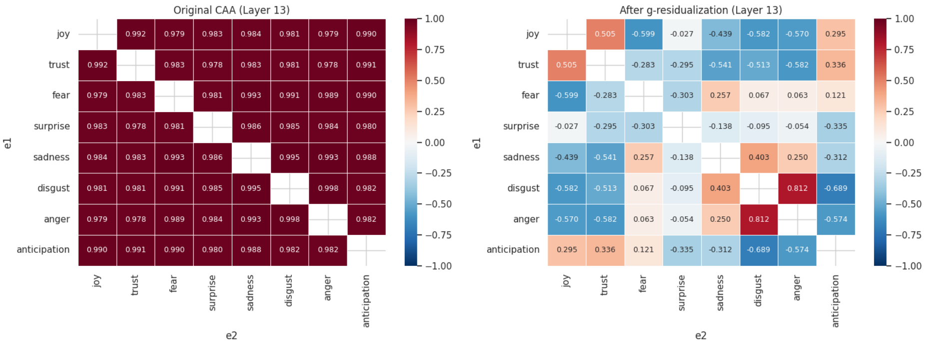

Figure 4.1: Pairwise cosine similarities among pooled CAA directions before (left) and after (right) removal of the shared emotionalisation component $\mathbf{g}$ at layer 13. Before residualisation, all directions are near-collinear. After residualisation, Plutchik opposite pairs become negative and the emotion-specific structure is revealed.

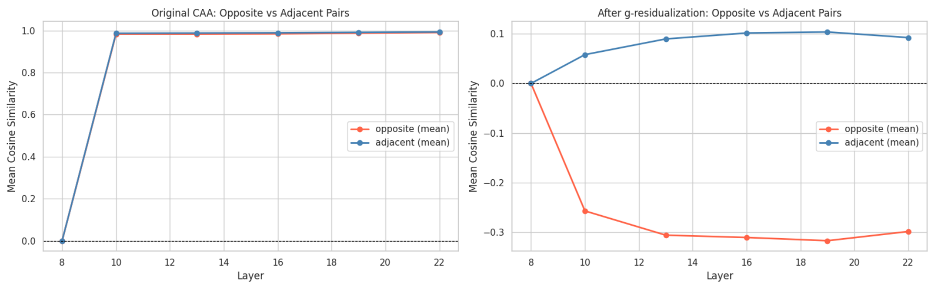

Figure 4.2: The average cosine similarity of opposing sentiment pairs and adjacent sentiment pairs in each layer before and after $\mathbf{g}$-residualisation. The tendency for the cosine values of opposing pairs to reverse from positive to negative is consistently observed in each layer.

It is also important to note that the residual space remains low-dimensional after this operation. The participation ratio changes only slightly from 3.61 to 3.63, and the $\mathrm{EVR}_{\mathrm{PC1}}$ for the eight $\mathbf{r}_e$ centre points is 0.448. This indicates that emotion-specific information is not scattered across a high-dimensional space, but is still largely organised within approximately four effective dimensions.

Individual residual shifts remain discriminative

The geometric structure of the centre-of-mass level, as described above, suggests that the residual vectors preserve emotion-specific structures. However, this structure alone does not enable us to determine whether individual emotional shifts consistently retain information about emotional labels. In principle, the observed structure could be the result of averaging a large number of noisy samples. To verify this, we trained a linear probe for each individual $\mathbf{g}$-residualised shift.

For each source utterance n and emotion e, we define the per-source affective shift as

$$\boldsymbol{\delta}_{n,e} = \mathbf{h}_{n,e}^{\text{emo}} - \mathbf{h}_{n}^{\text{neu}}.$$

We then remove the shared emotionalisation direction g:

$$\boldsymbol{\delta}_{n,e}^{g\perp} = \boldsymbol{\delta}_{n,e} - (\boldsymbol{\delta}_{n,e}^{\top}\mathbf{g})\mathbf{g}.$$

A linear probe, consisting of PCA-64 followed by standardised logistic regression, is trained to predict the emotion label e from $\boldsymbol{\delta}_{n,e}^{g\perp}$. We use five-fold cross-validation grouped by source identity.

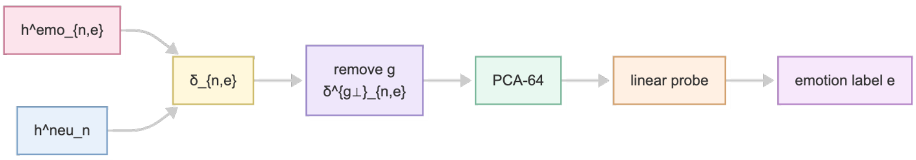

Figure 4.3: Flowchart of the linear probe analysis for the discriminability of $\hat{\mathbf{r}}_e$. The probe is trained on the g-residualised per-source affective shifts $\boldsymbol{\delta}_{n,e}^{g\perp}$ to predict emotion labels.

Layer	Balanced accuracy	Std
10	0.853	0.002
13 *	**0.857**	0.001
16	0.755	0.002

Table 4.2: Emotion-label balanced accuracy from a linear probe trained on $\boldsymbol{\delta}_{n,e}^{g\perp}$. Chance level is 0.125.

At layer 13, the balanced accuracy reaches 0.857, which is roughly seven times the chance level of 0.125. The g-residualised shifts therefore contain linearly decodable emotional category information that is independent of the shared emotionalisation component. This signal peaks around layers 10–13 and decreases at layer 16, suggesting that emotion-specific residual information is most explicit in the middle layers.

This result confirms that $\hat{\mathbf{r}}_e$ is not merely a geometric artefact of averaging. Individual residual shifts remain strongly label-informative.

Motivation for analysing internal variance

Together, these results show that the emotion-specific residual is meaningful at two levels. At the centroid level, the residual directions recover a Plutchik-like geometry after removal of the shared emotionalisation component. At the per-source level, individual residual shifts remain strongly emotion-discriminative.

However, a single residual direction $\mathbf{r}_e$ still summarises an entire distribution of affective shifts. Per-source analysis shows that the individual delta vectors $\boldsymbol{\delta}_{n,e}$ within each named emotion are not tightly concentrated around their centroid $\mathbf{r}_e$. In the local PCA within each emotion, the contribution ratio is approximately 3.5–3.7 for the top-5 component analysis, and the first local principal component explains only 11–12% of intra-emotion variance.

In other words, while $\mathbf{r}_e$ is a useful summary direction for emotion e, each emotion behaves more like a **cone** or a **prototype region** with internal sub-axes than a single narrow vector.

Consequently, the following section will examine whether these internal distributions are concentrated along a single prototypical axis, or span multiple axes. More broadly, we will consider whether

- the internal axes within each emotion category are shared across different emotions, and
- whether these axes correspond to interpretable, pre-verbal affective dimensions.

4.2 Internal Variance Structure of Named Emotions

The previous section showed that $\hat{\mathbf{r}}_e$ contains genuine emotion-specific information. However, $\hat{\mathbf{r}}_e$ is merely the average of many individual residual shifts $\boldsymbol{\delta}^{g\perp}_{n,e,i}$ for the same emotion. These individual shifts may either cluster tightly around the average direction, or spread across several distinct directions.

This section therefore examines the shape of this within-emotion distribution. Specifically, we ask whether each named emotion is well described by a single prototype direction, or whether it occupies a broader, multi-dimensional region in residual space.

Protocol

For each emotion e, we collect the per-source, per-intensity residuals $\boldsymbol{\delta}^{g\perp}_{n,e,i}$ at layer 13, apply PCA, and compute the participation ratio

$$\mathrm{PR}_e = \frac{\left(\sum_k \lambda_k\right)^2}{\sum_k \lambda_k^2},$$

where λ_k are the eigenvalues of the empirical covariance. To establish a baseline, we shuffle the emotion labels to destroy emotion-specific structure and report $\mathrm{PR}_{\text{shuffle}}$. A value $\mathrm{PR}_e < \mathrm{PR}_{\text{shuffle}}$ indicates that the emotion's internal distribution is more concentrated than random, whereas $\mathrm{PR}_e > \mathrm{PR}_{\text{shuffle}}$ indicates a more diffuse, multi-axis structure.

Results

Emotion	PR	PR (shuffle)	PR excess
joy	6.72	13.62	-6.90
trust	10.54	13.65	-3.11
sadness	12.11	13.59	-1.48
disgust	12.74	13.61	-0.87
anticipation	15.30	13.65	$+1.65$
anger	15.56	13.63	$+1.93$
fear	18.16	13.60	$+4.57$
surprise	18.94	13.63	$+5.31$

Table 4.3: Participation ratio of per-source affective shift distributions at layer 13. A negative excess indicates a near-uniaxial structure, and a positive excess indicates a multi-axis structure.

The results reveal a clear gradient. Focusing on joy and trust, it is evident that the PR values are significantly lower than the shuffle values. In other words, it can be said that the internal distribution for these two emotions is extremely concentrated. In contrast, the distributions for fear and surprise are quite broad (i.e., the PR values are significantly higher than the shuffle values).

These results demonstrate that the named emotions are not equivalent in their internal dimensions. Whilst some point to a narrow domain, others occupy a broader emotional landscape.

4.3 Local Sub-Axes and Their Affective Interpretation

The participation ratio analysis in the previous section shows that each named emotion has non-trivial internal variation. We now characterise this variation explicitly by extracting local principal axes.

Extraction pipeline

Starting from the per-source affective shifts, we apply the following residualisation steps before extracting emotion-internal principal components.

1. Compute the base shift:
$$\boldsymbol{\delta}_{n,e,i} = \mathbf{h}^{\text{emo}}_{n,e,i} - \mathbf{h}^{\text{neu}}_{n}$$

2. Remove the shared emotionalisation direction:

$$\boldsymbol{\delta}^{(1)}_{n,e,i} = \boldsymbol{\delta}_{n,e,i} - (\boldsymbol{\delta}^{\top}_{n,e,i}\mathbf{g})\mathbf{g}$$

3. Remove the per-emotion intensity direction $\mathbf{u}_e$ (the normalised mean of $\boldsymbol{\delta}^{(1)}_{n,e,\cdot}$ over intensities i):
$$\boldsymbol{\delta}^{(2)}_{n,e,i} = \boldsymbol{\delta}^{(1)}_{n,e,i} - (\boldsymbol{\delta}^{(1)}_{n,e,i}{}^{\top}\mathbf{u}_e)\mathbf{u}_e$$
4. Remove source-wise mean (mean over emotions for the same source n):
$$\boldsymbol{\delta}^{(3)}_{n,e,i} = \boldsymbol{\delta}^{(2)}_{n,e,i} - \frac{1}{|E|}\sum_{e'} \boldsymbol{\delta}^{(2)}_{n,e',i}$$

For each emotion e, we collect the matrix $\{\boldsymbol{\delta}^{(3)}_{n,e,i}\}_{n,i}$ and apply PCA. The k-th principal component $\mathbf{u}_{e,k}$ is the direction of maximal residual variance within emotion e that is orthogonal to $\mathbf{g}$ and to $\mathbf{u}_e$.

Interpretation of local sub-axes

We submit the top-5 and bottom-5 example texts for each $\mathbf{u}_{e,k}$ to an LLM judge (GPT-4o) and ask it to label the axis semantically. Table 4.4 summarises the leading axis for each emotion, along with judge-assigned contamination and confidence scores.

Emotion	**PC1 axis name**	**Contam.**	**Conf.**
joy	reflective/perspective-taking $\leftrightarrow$ immediate/external event	medium	4
trust	hopeful acceptance $\leftrightarrow$ self-assured confidence	medium	5
fear	immediate situational anxiety $\leftrightarrow$ chronic/existential worry	low	5
surprise	unexpected events $\leftrightarrow$ expected/anticipated events	low	5
sadness	loss & disappointment $\leftrightarrow$ everyday contentment	low	5
disgust	visceral sensory repulsion $\leftrightarrow$ mild/abstract discomfort	medium	4
anger	irritation/frustration $\leftrightarrow$ disappointment/resigned upset	low	5
anticipation	hopeful curiosity $\leftrightarrow$ uneasy waiting	low	5

Table 4.4: LLM-judge interpretations of the leading local sub-axis $\mathbf{u}_{e,1}$ for each emotion at layer 13. Five of eight emotions achieve `contamination = low` and `confidence = 5`, indicating that the extracted axes capture genuine affective distinctions rather than stylistic variation.

Style contamination check

To confirm that the local axes are affective rather than artefacts of text style or topic, we apply three independent checks to all 40 axes (8 emotions $\times$ 5 PCs).

1. **Style-feature probe**: We performed a regression analysis of the PC scores against a set of surface features (such as sentence length, lexical diversity and part-of-speech ratios). The average R^2 across the 40 features was 0.069, and only five features exceeded the threshold of 0.15.

2. **Neutral-control alignment**: We extracted PCA axes from the activation values of neutral paraphrases and measured the cosine similarity with each local principal component. The average maximum cosine value was 0.034, which is generally low, suggesting that there are no systematic confounding factors arising from stylistic variation at the paraphrase level.

3. **Meaning-preservation filter**: We restricted the analysis to the rewritten versions that passed through the semantic similarity filter and recalculated the PCA axes. The average cosine value between the full set of axes and the filtered axes was 0.944, and only one of the 40 axes ($\mathrm{surprise}$ PC5) fell below the robustness threshold of 0.70.

Based on these three checks, 34 of 40 axes are classified as likely affective, 6 as ambiguous, and none as likely stylistic. The local PCA axes therefore reflect genuine sub-distinctions within the affective content of each emotion category rather than co-varying surface features.

4.4 Cross-Emotion Sharing: From Local Axes to Meta-Axes

So far, the local sub-axes $\mathbf{u}_{e,k}$ have been characterised individually for each emotion. Here, we examine whether these axes are emotion-specific or whether they encode dimensions that recur across emotions. If sub-axes are shared, the internal variations within each emotion are organised along the same underlying dimensions, and $\hat{\mathbf{r}}_e$ can be reinterpreted as specific activation patterns along those shared dimensions. The correlation coefficient between s and the filtered axes was 0.944, and only one of the 40 axes ($\mathrm{surprise}$ PC5) fell below the robustness threshold of 0.70.

Clustering reveals PC-rank, not emotion label, as the organising principle

We calculated the absolute cosine similarity for all 40 local PC vectors (8 emotions $\times$ 5 PCs) and applied Ward's hierarchical clustering [30]. The resulting dendrogram revealed a surprising pattern: the main clusters were grouped not by emotion label, but based on the **rank** ($k = 1, 2, 3$) of the axes. Specifically:

- The $k = 1$ axes of all eight emotions form a single, tight cluster ($|\cos| \approx 0.74$–0.95 across all pairs).

- The $k = 2$ axes of all eight emotions form a second cluster.
- The $k = 3$ axes similarly form a third cluster.
- Emotion-specific structure first appears at $k = 4$ and $k = 5$.

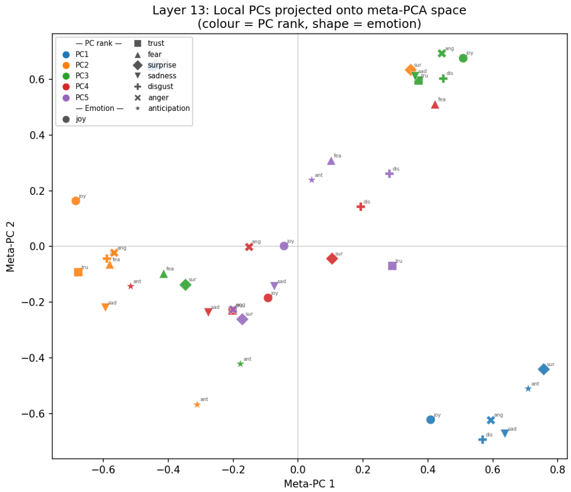

Figure 4.4: PCA scatter plot of the 40 local PC vectors $\mathbf{u}_{e,k}$ at layer 13. The first two meta-components from a PCA over the 40 vectors are plotted, with points coloured by PC rank and shaped by emotion label. The clustering is primarily determined by PC rank rather than emotion label, indicating that the leading local axes capture shared affective dimensions across all emotions.

This organisation is confirmed by cross-emotion cosine similarities at the PC1 level (Table 4.5) and by meta-PCA over the 40 vectors. The top-3 meta-components capture 51% of the total variance across all emotion-PC combinations and separate by rank.

Emotion pair (PC1)	$\lvert\cos(\mathbf{u}_{e_1,1}, \mathbf{u}_{e_2,1})\rvert$
disgust – anger	0.950
sadness – disgust	0.922
trust – disgust	0.886
joy – anger	0.874
fear – disgust	0.839
anticipation – sadness	0.805
surprise – sadness	0.743

Table 4.5: Cross-emotion absolute cosine similarity for the leading local sub-axis $\mathbf{u}_{e,1}$. All pairs are well above chance, indicating a universal primary axis shared across all eight named emotions.

Formalisation of meta-axes $\mathbf{m}_k$

Given that k-th rank local axes cluster tightly across emotions, we define a shared **meta-axis** $\mathbf{m}_k$ as the normalised sign-aligned mean of the k-th local PC vectors:

$$\mathbf{m}_k = \frac{\sum_{e \in E} \sigma_{e,k} \, \mathbf{u}_{e,k}}{\left\| \sum_{e \in E} \sigma_{e,k} \, \mathbf{u}_{e,k} \right\|_2},$$

where $\sigma_{e,k} \in \{+1, -1\}$ is chosen by sign-aligning $\mathbf{u}_{e,k}$ with the first emotion's axis so that all vectors point in a consistent direction before averaging.

As a robustness check, we also extract $\mathbf{m}_k$ via group PCA (applying PCA to the sign-aligned set of k-th rank vectors) and measure the cosine between the two estimates.

Meta-axis	$\lvert\cos(\mathbf{m}_k^{\mathrm{mean}}, \mathbf{m}_k^{\mathrm{pca}})\rvert$	**Mean** $\lvert\cos\rvert$ **to** $\mathbf{u}_{e,k}$
$\mathbf{m}_1$	0.990	0.892
$\mathbf{m}_2$	0.829	0.800
$\mathbf{m}_3$	0.989	0.672
$\mathbf{m}_4$	0.266	0.437
$\mathbf{m}_5$	0.104	0.427

Table 4.6: Stability of meta-axes $\mathbf{m}_k$. The first column reports agreement between the two extraction methods (sign-aligned mean vs. group PCA). The second reports mean absolute cosine between $\mathbf{m}_k$ and the individual local axes $\mathbf{u}_{e,k}$. Meta-axes $\mathbf{m}_1$–$\mathbf{m}_3$ are stable and representative, while $\mathbf{m}_4$ and $\mathbf{m}_5$ are not.

Meta-axes $\mathbf{m}_1$, $\mathbf{m}_2$, and $\mathbf{m}_3$ are stable under method variation (method agreement ≥ 0.83) and well-aligned with the individual local axes (mean $\lvert\cos\rvert \geq 0.67$). Meta-axes $\mathbf{m}_4$ and $\mathbf{m}_5$ show poor method agreement (0.27 and 0.10) and are excluded from further interpretation.

Stable meta-axes have two further notable characteristics. Firstly, all $\mathbf{m}_k$ are nearly orthogonal to the common emotionalisation direction $\mathbf{g}$ (for all k, $\lvert\cos(\mathbf{m}_k, \mathbf{g})\rvert < 0.02$). This supports the notion that they capture the **emotional texture** rather than the overall strength of emotionalisation. Secondly, all $\mathbf{m}_k$ are nearly orthogonal to the steering residuals $\hat{\mathbf{r}}_e$ of each named emotion (i.e., for all combinations, $\lvert\cos(\mathbf{m}_k, \hat{\mathbf{r}}_e)\rvert < 0.09$). This implies that the meta-axes cannot be reconstructed from the averaged prototypes alone. Instead, they encode the intra-emotion variance that is obscured by the averaging process.

4.5 Pre-Verbal Interpretation of Meta-Axes

We now examine the emotional dimensions represented by each meta-axis, $\mathbf{m}_k$. The meta-axes are derived from aggregated activations across all named emotions. Their semantic content can therefore be described independently of specific emotion labels. To interpret these axes, we presented the LLM evaluator with examples of extreme activations extracted from all eight emotions simultaneously. The evaluator was instructed to assign distinguishable labels to the five axes. These labels had to describe the factors that separate the high and low poles across emotion categories.

This experiment differs from the evaluation in Section 4.3, because in Section 4.3 we examined nuances within individual emotions whereas this experiment aims to reveal dimensions that are more fundamental than emotion labels.

	Axis name	**Pre-verbal interpretation**	**Contam.**	**Conf.**
$\mathbf{m}_1$	Agitated/reactive **vs** mellow/accepting	Arousal / affective reactivity: how strongly and immediately a state is felt	medium	5
$\mathbf{m}_2$	Nostalgic/ruminative **vs** pragmatic/forward-looking	Temporal orientation: backward-looking rumination vs. present-focused pragmatic coping	medium	4
$\mathbf{m}_3$	Interpersonally engaged / closure-seeking **vs** self-contained / processing	Social vs. internal orientation: whether affect is directed outward or processed internally	medium	4
$\mathbf{m}_4$	Mundane/obligatory **vs** meaningful/fulfilling	Existential weight: trivial routine vs. personally significant experience	medium	4
$\mathbf{m}_5$	Relief/release **vs** tension/constraint	Somatic pressure: physical or situational release vs. constriction	medium	5

Table 4.7: LLM-judge interpretation of the five meta-axes at layer 13. Labels are derived from contrastive evaluation across all eight named emotions simultaneously, yielding emotion-independent descriptions.

These five dimensions correspond to established conceptual frameworks in the psychology of emotion. Specifically, $\mathbf{m}_1$ corresponds to the arousal dimension in Russell's Circumplex Model [31], and $\mathbf{m}_2$ corresponds to temporal orientation in rumination theory [32]. $\mathbf{m}_3$ corresponds to the social sharing of emotions [33], $\mathbf{m}_4$ relates to the evaluation of relevance and importance in evaluation theory [34], and $\mathbf{m}_5$ is similar to Damasio's somatic marker hypothesis [35]. None of these dimensions are emotion-specific. Each spans all eight named categories and can vary independently of the emotion label.

4.6 Re-Interpreting $\hat{\mathbf{r}}_e$ as a Pre-Verbal Projection

The experiments so far establish that

1. each named emotion has internal sub-axes
2. these sub-axes are shared across emotion categories
3. the shared meta-axes have pre-verbal, emotion-independent interpretations

This section draws these threads together into a revised decomposition of $\hat{\mathbf{r}}_e$.

Decomposition

Each meta-axis $\mathbf{m}_k$ defines a direction in residual space. For a given emotion e, we define the **projection coefficient**

$$\beta_{e,k} = \hat{\mathbf{r}}_e^\top \mathbf{m}_k$$

and write the decomposition

$$\hat{\mathbf{r}}_e = \underbrace{\sum_{k=1}^{K} \beta_{e,k}\, \mathbf{m}_k}_{\text{pre-verbal projection}} + \underbrace{\boldsymbol{\varepsilon}_e}_{\text{emotion-specific residual}},$$

where $\boldsymbol{\varepsilon}_e \perp \mathbf{m}_k$ for all k, and $K = 5$ (all extracted meta-axes, since $\mathbf{m}_1$–$\mathbf{m}_3$ are stable while $\mathbf{m}_4$, $\mathbf{m}_5$ have lower method agreement). The explained variance ratio is $\mathrm{EVR}(e) = \sum_{k=1}^{K} \beta_{e,k}^2$ (since $\hat{\mathbf{r}}_e$ is a unit vector).

Results

Emotion	$\beta_{e,1}$	$\beta_{e,2}$	$\beta_{e,3}$	$\beta_{e,4}$	$\beta_{e,5}$	**EVR**
joy	$+0.160$	-0.261	-0.073	$+0.156$	$+0.269$	0.238
trust	-0.062	-0.249	$+0.152$	-0.211	$+0.235$	0.185
fear	-0.442	$+0.060$	$+0.162$	-0.073	-0.102	0.231
surprise	$+0.224$	$+0.553$	-0.256	-0.083	-0.043	0.399
sadness	$+0.070$	$+0.174$	-0.089	$+0.429$	-0.227	0.263
disgust	-0.041	$+0.041$	-0.151	$+0.066$	-0.295	0.125
anger	-0.034	-0.008	$+0.035$	-0.060	-0.242	0.058
anticipation	$+0.050$	-0.319	$+0.236$	-0.083	$+0.247$	0.227
Mean						**0.216**

Table 4.8: Projection coefficients $\beta_{e,k} = \hat{\mathbf{r}}_e^\top \mathbf{m}_k$ and explained variance ratio (EVR) across all five meta-axes.

The $\beta_{e,k}$ pattern constitutes the "affective fingerprint" of each named emotion in the pre-verbal meta-axis space. All five meta-axes together explain approximately 22% of the variance in $\hat{\mathbf{r}}_e$, while the remaining 78% is the emotion-specific residual $\boldsymbol{\varepsilon}_e$.

Several interpretable patterns can be observed in the $\beta_{e,k}$ coefficients. Fear has the most negative value, $\beta_{e,1}$ (-0.442), and is clearly situated at the "calm/objective" pole of the arousal meta-axis. This is consistent with the view that fear is not a state of high arousal, but rather a state of vigilant inhibition. Surprise has the largest positive value of $\beta_{e,2}$ ($+0.553$) and is located at the ruminative end of the temporal orientation axis. This reflects the retrospective nature of processing unexpected events. Sadness exhibits the largest positive value of $\beta_{e,4}$ ($+0.429$), corresponding to the 'meaningful/fulfilling' end of the existential weight axis. This is consistent with the perception of loss as something possessing personal significance. Expectation exhibits the most negative value, $\beta_{e,2} = -0.319$, and is situated at the practical/future-oriented pole. This contrasts with the retrospective orientation of surprise despite both involving uncertainty. Joy is characterised primarily by $\beta_{e,5} = +0.269$ (release/relief) and $\beta_{e,2} = -0.261$ (forward-looking), reflecting a state of positive resolution rather than turmoil. Trust is divided into $\beta_{e,2} = -0.249$ (future-oriented) and $\beta_{e,4} = -0.211$ (routine/obligatory), suggesting that trust is encoded as practical, low-risk acceptance rather than something of deep personal significance.

Connecting to Chapter 3

We can now state the revised decomposition at the level of the steering vector. Chapter 3 used the unit-normalised residual $\hat{\mathbf{r}}_e$ as the primitive emotion-specific direction, namely that:

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e. \qquad \text{(Chapter 3)}$$

Chapter 4 decomposes $\hat{\mathbf{r}}_e$ into a pre-verbal projection and an orthogonal emotion-specific residual:

$$\hat{\mathbf{r}}_e = \sum_{k=1}^{K} \beta_{e,k}\, \mathbf{m}_k + \boldsymbol{\varepsilon}_e, \qquad \beta_{e,k} = \hat{\mathbf{r}}_e^{\top} \mathbf{m}_k, \qquad \boldsymbol{\varepsilon}_e \perp \mathbf{m}_k. \qquad \text{(Chapter 4)}$$

Substituting into the steering formula:

$$\Delta_e = \alpha_G \mathbf{g} + \underbrace{\alpha_R \sum_{k=1}^{K} \beta_{e,k}\, \mathbf{m}_k}_{\text{pre-verbal component}} + \underbrace{\alpha_R \boldsymbol{\varepsilon}_e}_{\text{emotion-specific residual}}. \qquad \text{(Combined)}$$

The practical question is whether the $\alpha_R \boldsymbol{\varepsilon}_e$ term contributes useful signal or noise. This is answered directly in Section 4.8. Condition A steers with the full $\alpha_R \hat{\mathbf{r}}_e$ (which includes $\boldsymbol{\varepsilon}_e$), while Condition B steers with only $\alpha_R \sum_k \beta_{e,k} \mathbf{m}_k$ (which excludes it). Their comparison is therefore a direct empirical test of $\boldsymbol{\varepsilon}_e$'s functional role.

In sum, what Chapter 3 called $\hat{\mathbf{r}}_e$ is not a primitive atom. Named emotions are reinterpreted as **specific patterns of activation** along label-free pre-verbal dimensions $\mathbf{m}_k$, encoded by the fingerprint $(\beta_{e,1}, \ldots, \beta_{e,K})$ in Table 4.8.

4.7 Causal Validation: Local Axis Steering

The internal axes identified by PCA are geometric features of the activation distribution. To establish that they are also causally relevant to generation, we directly steer along them and measure whether the outputs shift in the expected affective direction.

Steering formula

We extend the Chapter 3 intervention to include a local axis component:

$$\Delta_e(\alpha_G, \alpha_R, \beta) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e \pm \beta\, \mathbf{u}_{e,1},$$

where $+\beta$ steers toward the high pole of the axis and $-\beta$ toward the low pole. In a parallel condition, we substitute the shared meta-axis $\mathbf{m}_1$ for the emotion-specific $\mathbf{u}_{e,1}$.

Results

Emotion	n	`local_axis_match`	`target_match`	`fluency`	`style_contam.`
anger	48	**0.445**	0.242	0.906	0.667
anticipation	48	0.305	0.411	0.844	0.631
fear	48	0.307	0.388	0.842	0.626
joy	48	0.279	0.405	0.790	0.591
sadness	48	0.258	0.340	0.884	0.630
surprise	48	0.261	0.280	0.762	0.566
trust	48	0.184	0.259	0.814	0.610
disgust	48	0.163	0.213	0.883	0.654

Table 4.9: LLM-judge evaluation of local-axis steering ($\pm\beta\, \mathbf{u}_{e,1}$) at layer 13 (Exp 5). Eight emotions, 24 seed texts, 2 poles. `local_axis_match` is the judge's score for the intended sub-axis shift, and `target_match` is the overall emotion match. Fluency remains high throughout.

Steering along the local axis produces detectable affective shifts for most emotions, with anger achieving the strongest `local_axis_match` of 0.445 despite using a conservative $\beta = 0.35$. Additionally, an important asymmetry emerges at the pole level. For anger, the low pole (calm/resigned) produces a local axis match of 0.602, whereas the high pole (agitated/reactive) yields only 0.289. This suggests that the base model's generation tendencies are asymmetrically distributed along the meta-axis, a point revisited in Section 4.8.

When the emotion-specific $\mathbf{u}_{e,1}$ is replaced by the shared $\mathbf{m}_1$, overall performance is comparable (target match: 0.923 vs 0.926, local axis match: 0.519 vs 0.525), demonstrating that the shared meta-axis captures the essential causal component of the local sub-axis.

4.8 Label-Free Affective Steering

The Combined formula in Section 4.6 isolates a specific empirical question. Is ε_e (the 78% of $\hat{\mathbf{r}}_e$ that cannot be expressed via $\mathbf{m}_k$) functionally useful for steering, or is it inert noise? This experiment is designed to answer that question directly by constructing three teering conditions that differ only in whether and how ε_e is included.

Three steering conditions

We compare three steering conditions using the same coefficient magnitudes:

A: $\Delta = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$

Since $\hat{\mathbf{r}}_e = \sum_k \beta_{e,k} \mathbf{m}_k + \varepsilon_e$, this steers with **both** the pre-verbal projection and ε_e. This is equivalent to the Chapter 3 intervention.

B: $\Delta = \alpha_G \mathbf{g} + \sum_k \tilde{\beta}_{e,k}\, \mathbf{m}_k$ where $\tilde{\beta}_{e,k} = \alpha_R \beta_{e,k} = \alpha_R(\hat{\mathbf{r}}_e^\top \mathbf{m}_k)$

This steers with only the $\mathbf{m}_k$-projection of $\hat{\mathbf{r}}_e$, **explicitly excluding** ε_e. The difference A−B therefore isolates the causal contribution of ε_e.

C: $\Delta = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e + \sum_k \tilde{\beta}_{e,k}\, \mathbf{m}_k$

which adds the meta-axis projection on top of the full named prototype. Since $\hat{\mathbf{r}}_e$ already contains the $\mathbf{m}_k$ component, this condition double-weights the pre-verbal projection relative to ε_e.

Results

Metric	A	B	C	B−A	C−A
target emotion match	0.936	**0.938**	0.935	$+0.002$	-0.001
meaning preservation	0.881	**0.902**	0.879	$+0.021$	-0.002
fluency	0.820	**0.891**	0.793	$+0.071$	-0.027
style contamination	0.699	**0.735**	0.690	$+0.036$	-0.009
subtle affective state	0.712	**0.723**	0.686	$+0.011$	-0.026
label-free nuance	0.678	**0.697**	0.650	$+0.019$	-0.028

Table 4.10: Comparison of three steering conditions on 24 seed texts at layer 13 (Exp 10). Condition B (label-free $\mathbf{m}_k$ projection) matches or exceeds Condition A (named emotion prototype) on all metrics, with a particularly notable improvement in fluency ($+0.071$). Condition C (hybrid) falls below both A and B, indicating that adding ε_e to the meta-axis projection reduces quality rather than enhancing it.

Condition B achieves the same target-emotion match as Condition A (0.938 vs 0.936) while producing noticeably more natural outputs (fluency: 0.891 vs 0.820). Condition C, which adds the full $\hat{\mathbf{r}}_e$ on top of the meta-axis projection, is worse than both Condition A and Condition B on every metric. This pattern implies that the ε_e component does not provide additional affective nuance, but introduces noise that degrades output fluency.

The emotion-wise breakdown reveals one important exception: surprise degrades under Condition B (fluency -0.180, style contamination -0.213), consistent with the earlier observation that surprise has the highest participation ratio (PR excess $+5.31$) and the lowest alignment between its local axes and the shared meta-axes. This suggests that surprise has genuine emotion-specific structure in its ε_e component that is not captured by $\mathbf{m}_k$.

This results support the refined view of emotion representation, that named emotion labels function as convenient short-hands for specific activation **patterns** in the pre-verbal meta-axis space. The $\beta_{e,k}$ values in Table 4.8 are the core content of the emotion label. Also, the named prototype $\hat{\mathbf{r}}_e$ is largely recoverable from these projections, and the residual ε_e is mostly non-functional for generation. Steering with $\sum_k \beta_{e,k}\mathbf{m}_k$ is therefore not only theoretically cleaner but also empirically superior in terms of output quality.

4.9 Limitations and Transition

The experiments in this chapter revealed that the emotion-specific residual $\hat{\mathbf{r}}_e$, derived in Chapter 3, possesses structured internal geometric properties. Its principal variance is organised along a common meta-axis $\mathbf{m}_k$, which has a label-independent interpretation that predates verbalisation. The decomposition formula $\hat{\mathbf{r}}_e = \sum_k \beta_{e,k}\mathbf{m}_k + \varepsilon_e$ extends the equation from Chapter 1 into a three-tiered hierarchy: shared affective components ($\mathbf{g}$), pre-linguistic affective dimensions ($\mathbf{m}_k$), and emotion-specific small residuals (ε_e).

The experiments in this chapter revealed that the emotion-specific residuals $\hat{\mathbf{r}}_e$, derived in Chapter 3, possess structured internal geometric properties. Their principal variance is organised along a common meta-axis $\mathbf{m}_k$, for which a label-independent interpretation exists prior to verbalisation. The decomposition formula $\hat{\mathbf{r}}_e = \sum_k \beta_{e,k}\mathbf{m}_k + \varepsilon_e$ extends the equation from Chapter 3 into a three-tier hierarchical structure:

1. Shared emotional components ($\mathbf{g}$)
2. Pre-verbal emotional dimensions ($\mathbf{m}_k$), and
3. Emotion-specific small residuals (ε_e).

These conclusions have several limitations.

Firstly, the projection onto the meta-axis explains only approximately 22% of the variance in $\hat{\mathbf{r}}_e$, with the remaining 78% accounted for by ε_e. The label-free steering experiment (Section 4.8) provides empirical evidence that, for most emotions, ε_e contributes as noise rather than signal; however, this does not imply that ε_e is semantically empty. ε_e may encode relevant emotion-specific information under different generative conditions, at different coefficient scales, or with regard to specific emotions (particularly surprise, which is the only case where Condition A outperforms Condition B). The current evidence merely indicates that ε_e is functionally impotent “under the tested conditions”; it does not demonstrate that it is inherently meaningless.

Secondly, the LLM judgement labels for $\mathbf{m}_k$ are assigned a “moderate” level of contamination, reflecting the inherent difficulty of generalisation across emotions. Although this interpretation is plausible, it has not been independently verified by human evaluators or psychometric tools. We also need to note that the meta-axes $\mathbf{m}_4$ and $\mathbf{m}_5$ are unstable (inter-method agreement < 0.3) and are included only in the calculation of EVR. They are not used as interpretable axes.

Thirdly, this evidence is based on a single model (Llama-3.1-8B-Instruct) and a single layer depth (13 layers). Further investigation is required regarding the generalisability to other models and layer depths within the meta-axis structure.

These limitations provide the motivation for the next chapter. The label-free $\mathbf{m}_k$ axes provide a theoretical foundation for our Emotion Engine user interface that is not bound by fixed emotion labels. In the next chapter, we consider how such pre-verbal representations can be made operational in practical user interfaces, and whether the separation of emotionalisation vector ($\mathbf{g}$), emotion texture ($\mathbf{m}_k$), and emotion-specific identity (ε_e) can be transformed into meaningful and controllable operational elements for users.

Chapter 5

Evaluation

This chapter describes how the steering behaviours identified in the previous chapter will be evaluated in a controlled and human-centred manner. The goal here is not to re-establish the existence of affective structure in residual space, but to assess whether the extracted control directions yield perceptible, monotonic, and semantically stable emotional edits.

5.1 Evaluation Setup

We prepare a set of input texts spanning neutral descriptions, short narratives and conversational utterances. For each input, the system first extracts a baseline emotion vector, which is then modified along one or more predefined directions.

For controlled experiments, we consider three conditions:

1. no steering (baseline)
2. single dimensional manipulation within the PAD space, where users adjust pleasure, arousal, or dominance
3. two-dimensional manipulations within the PAD space

We define three primary metrics to evaluate the behaviour of the emotion code:

1. **Shift Accuracy**: the proportion of cases in which human annotators agree that the steered output expresses a stronger presence of the target emotion than the baseline
2. **Intensity Monotonicity**: evaluates whether increasing the magnitude of a steering vector leads to a monotonic increase in perceived emotional intensity. This metric directly tests whether emotion behaves as a continuous and scalable latent variable.

3. **Content Stability**: whether semantic content is preserved under emotional steering. This is measured using a combination of human judgement and embedding-based semantic similarity in concept space.

5.2 Human-Centred Evaluation

In addition to controlled steering experiments, we would like to conduct a small-scale user study focusing on interactive manipulation of emotion vectors. Participants are presented with the PAD visualisation and are asked to modify the emotional state of a given input text to match an intended affective description (e.g. "more tense but less negative").

We evaluate whether users can reliably reach similar regions of the PAD space for similar intentions, whether users report that changes in the output align with their intuitive expectations, and whether fine-grained adjustments lead to perceptible but non-disruptive changes in tone. This evaluation focuses on intuitive usability rather than task performance, reflecting the project's goal of enabling pre-verbal emotional editing rather than emotion classification.

5.3 Failure Cases and Limitations

For the cases where steering fails to produce the intended emotional shift or results in semantic distortion, we would like to analyse how those happened. It might be the case that the steering magnitudes are too high for the humans to recognise the differences, or the interference between emotion dimensions. Such cases are not treated as errors but as signals of the limits of linear emotion composition and of the PAD projection itself, hence better ways of representing the latent emotion representation should be further investigated.

Chapter 6

Conclusions and Future work

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